

Renal Handling of Organic Solutes

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CHAPTER OUTLINE

GLUCOSE, 225

AMINO ACIDS, 244

ORGANIC CATIONS AND ANIONS, 236

KEY POINTS

- In a normoglycemic adult, the kidneys filter 160–180 g/day of glucose (~30% of daily energy expenditure), which is reabsorbed by the sodium glucose cotransporter SGLT2 (~97%) and SGLT1 (~3%) in the early and late proximal tubule, respectively.
- The basal overall glucose tubular reabsorption capacities for SGLT2 versus SGLT1 are in the range of 3:1 to 5:1. The transport capacity of tubular SGLT1 is unmasked (up to ~80 g/day) when more glucose is delivered to the late proximal tubule (e.g., in diabetes or with SGLT2 inhibition).
- Inhibitors of SGLT2 are used as antihyperglycemic drugs that target renal glucose reabsorption, induce urinary glucose loss, and have clinical efficacy regarding lowering blood glucose. Even more importantly, they have protective effects on the kidney and heart through mechanisms that go beyond blood glucose lowering.
- One of the major roles of organic anionic transporters, long considered mainly drug and toxin transporters, now appears to be to regulate many aspects of endogenous physiology.
- OAT1 and OAT3 appear to be the major renal basolateral transporters involved in the elimination of numerous uremic toxins originating in the gut microbiome, although OCT2 is likely the main route of TMAO elimination.
- Dysregulation of renal uric acid /urate excretion leads to hyperuricemia and gout and is a risk factor for cardiometabolic diseases. The heritability of serum urate levels has led to the identification of the key urate transporters of the kidney.
- The Remote Sensing and Signaling Theory is a systems biology theory about the role of SLC and ABC transporters in the interorgan and interorganismal (e.g., host-gut microbiome) “remote” communication via transporter-mediated movement of metabolites, signaling molecules, gut microbiome products, nutrients, uric acid, and uremic toxins into different body tissues and fluid compartments.
- Amino acid transporters often form heterodimers with ancillary subunits that are essential for trafficking of the transporters to the cell surface. Genetic complexity is observed in renal aminoacidurias due to heterodimer formation and transporter redundancy.
- Rare inherited aminoacidurias define four major routes of amino acid reabsorption. Amino acid antiporters play an important role in the apical and basolateral transport of cationic and neutral amino acids.

The kidneys must handle a multitude of organic solutes. Some are energy substrates or building blocks, like glucose or amino acids. The body does not want to lose these organic substrates, and they are normally not excreted into the urine. Considering a glomerular filtration rate (GFR) of ~180 L/

day in a healthy adult and concentrations in the plasma of many of these energy substrates in the mM or high μ M range, large amounts are filtered by the glomeruli and subsequently reabsorbed in the tubular system of the kidney, primarily in the proximal tubule. The tubular transport capacity for these

organic solutes is multiple-fold higher than their normal rate of glomerular filtration to ensure quantitative reabsorption. In contrast, organic anions and cations include many small-molecule drugs, metabolites, and toxins that the body needs to eliminate. To maximize their excretion, many of these substrates are filtered by the glomeruli and, in addition, are also secreted into the urine by a transtubular secretion system, primarily in the proximal tubule. Many of these compounds have a high-protein binding, such that their glomerular filtration is low and their excretion depends on tubular secretion. Although this tubular secretion system transports many drugs (e.g., antibiotics, antivirals, nonsteroidal antiinflammatory drugs [NSAIDs], and diuretics) and toxins (e.g., organic mercurials and aristolochic acid), it is now clear that this system plays a central role in local and systemic physiology in that it transports many endogenous metabolites, signaling molecules, vitamins, gut microbiome and dietary products, indicating a critical role beyond a simple renal excretory function.

This chapter summarizes the physiology and cell and molecular biology of the organic solute transport system in the kidney and the implications for body homeostasis. It also outlines clinically relevant aspects and the role of the system in genetic and metabolic disorders and as a therapeutic target.

GLUCOSE

The kidneys are a major site of glucose handling. This includes the continuous glomerular filtration of large amounts of glucose, almost all of which is subsequently reabsorbed by the proximal tubule, such that the formed urine in a healthy individual is nearly glucose free. The glucose reabsorbed by the proximal tubule is primarily taken up into the peritubular capillaries and provided as an energy source to further distal tubular segments or returned to the systemic circulation. Moreover, in addition to the liver, the kidneys contribute to the endogenous production of glucose or gluconeogenesis. Thus the kidneys use glucose as fuel but also contribute to maintaining blood glucose levels and overall metabolic balance by reabsorbing filtered glucose and generating new glucose. This is relevant in healthy individuals, particularly in the fasting state, and becomes pathophysiologically important in diabetes and hyperglycemic conditions. As a consequence, new antihyperglycemic drugs have been developed that target renal glucose reabsorption, induce urinary glucose loss, and have clinical efficacy with regard to lowering blood glucose. Even more importantly, they have protective effects on the kidney and cardiovascular system (see [Chapter 41](#) for more on diabetes and kidney disease).

PHYSIOLOGY OF RENAL GLUCOSE TRANSPORT

In many organisms including humans, the cellular uptake and metabolism of D-glucose serves as an important energy source.^{1,2} The brain primarily runs on glucose and depends on its continuous uptake, which alone requires ~125 g of glucose every day. As a consequence, glucose homeostasis is finely regulated and blood glucose levels are maintained in a range of 4–9 mmol/L by various hormones, including insulin and glucagon, that regulate glucose uptake into

target cells, as well as glucose storage and endogenous glucose production.^{1,2}

Glucose is a small molecule that is freely filtered by the glomeruli of the kidneys. Under conditions of normal blood glucose levels (~5.5 mmol/L or 100 mg/dL) and normal glomerular filtration rate (GFR, ~180 L/day), the kidneys filter 160–180 g of glucose each day. This is equal to ~30% of the daily energy expenditure, which would be lost into the urine if not regained by the renal tubules. Instead, more than 99% of the filtered glucose is reabsorbed by the tubules, primarily in the proximal tubule ([Fig. 8.1](#)). As described in more detail later, glucose reabsorption in the proximal tubule involves the two Na⁺-glucose cotransporters SGLT2 and SGLT1, which are expressed in the brush border membrane of the early and late proximal tubule, respectively. Na⁺-glucose cotransport is a saturable process and has a maximum transport capacity (T_{max}). The T_{max} of the kidneys for glucose can vary among individuals and averages ~430 g and ~500 g/day (300 and 350 mg/min) in female and male healthy human subjects, respectively.^{1,2} This equals ~threefold the normal tubular glucose load of 160–180 g/day so that the renal glucose reabsorption capacity is not saturated under conditions of normal blood glucose levels and GFR. Theoretically, at a normal GFR, the T_{max} should be reached and glucose should begin to be excreted in the urine at a plasma glucose threshold of ~15.5 mmol/L (280 mg/dL). The T_{max} for glucose of individual nephrons is variable, however, and thus low-level urinary glucose spilling begins in a normal, glucose-tolerant individual at modestly elevated plasma glucose levels of ~10–11 mmol/L (180–200 mg/dL; see “Splay,” [Fig. 8.2](#)). A robust and linear increase in urinary glucose excretion occurs when blood glucose levels rise above 15–16 mmol/L. GFR is a determinant of the filtered glucose load, and as a consequence, glucosuria can occur at lower plasma glucose concentrations when GFR is elevated (e.g., in pregnancy or in diabetes), or at higher blood glucose levels when GFR is reduced (e.g., in chronic kidney disease [CKD]). In addition, changes in the transport activity and expression level of SGLT2 and SGLT1 (see later) are expected to further modify this relationship.

A PRIMARY ROLE FOR SGLT2 IN RENAL GLUCOSE REABSORPTION

Experiments performed on isolated nephron segments of rabbit kidneys in the early 1980s identified differences in the rate of uptake and affinity for glucose between the early and late proximal tubule segments, respectively.³ Subsequent studies confirmed that the heterogeneity in glucose transport across the proximal tubule was attributed to the presence of two different glucose transporters in the brush border membrane.⁴ These studies and transport studies in membrane vesicles and analyses of mRNA expression in isolated nephron segments of rat and rabbit kidneys, as well as the cloning of the responsible genes, performed largely between 1981 and 1995, identified the Na⁺-glucose cotransporters SGLT2 (SLC5A2) and SGLT1 (SLC5A1) as the primary genes and pathways for renal glucose reabsorption.^{3–12} These studies established the concept that the “bulk” of tubular glucose uptake across the apical membrane occurs in the “early” proximal tubule (S1/2 segment) and is mediated by the low-affinity and

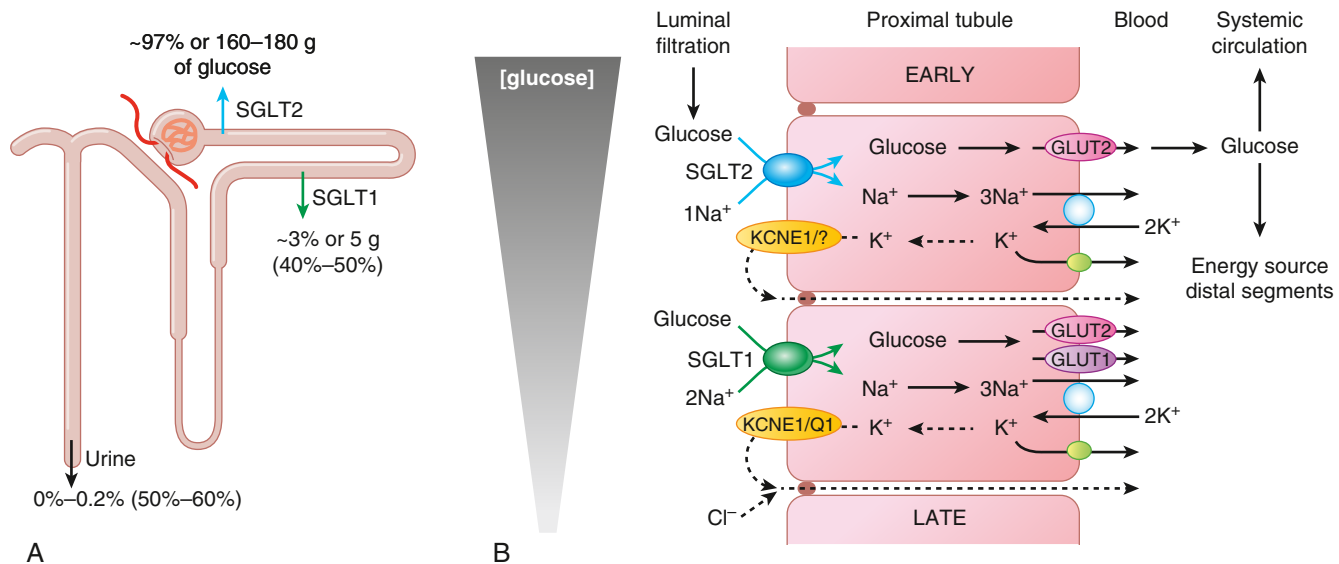


Fig. 8.1 Glucose transport in the kidney. (A) Under normoglycemia, ~97% of filtered glucose is reabsorbed via SGLT2 in the early segments of the proximal tubule. The remaining ~3% of glucose is reabsorbed by SGLT1 in the late proximal tubule such that the urine is nearly free of glucose. SGLT2 inhibition shifts glucose reabsorption downstream and unmasks the capacity of SGLT1 to reabsorb glucose (~40% of filtered glucose, depending on glucose load; see numbers in parentheses). (B) Cell model of glucose transport: The basolateral Na⁺-K⁺-ATPase lowers cytosolic Na⁺ concentrations and generates a negative interior voltage, thereby providing the driving force for Na⁺-coupled glucose uptake through SGLT2 and SGLT1 across the apical membrane. The facilitative glucose transporters GLUT2 and GLUT1 mediate glucose transport across the basolateral membrane down its chemical gradient. Na⁺-glucose cotransport is electrogenic and accompanied by paracellular Cl⁻ reabsorption or transcellular K⁺ secretion to stabilize membrane potential; K⁺ channels KCNE1/unknown α subunit and KCNE1/KCNQ1 in early and late proximal tubule, respectively. (Modified with permission from Vallon V. Molecular determinants of renal glucose transport. *Am J Physiol Cell Physiol.* 2011;300:C6–C8.)

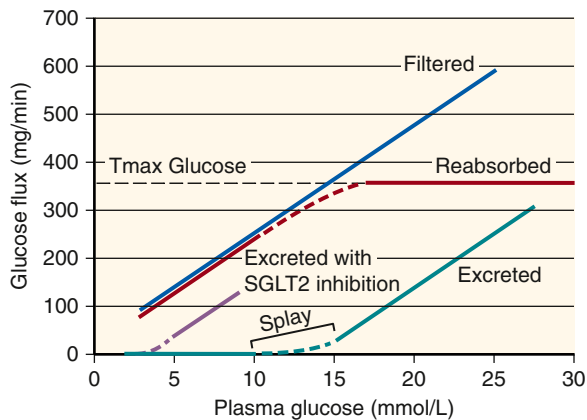


Fig. 8.2 Tubular reabsorption and urinary excretion of glucose as a function of filtered glucose. Tubular reabsorption of glucose (red) increases linearly with the filtered glucose load (blue) until reabsorption reaches the tubular reabsorption capacity (*T_{max} glucose*) and glucose starts to be excreted in the urine (green). SGLT2 inhibition reduces the renal glucose reabsorption to the transport capacity of SGLT1 and shifts the function to the left (purple)—that is, it reduces the renal glucose threshold (~3 mM) and *T_{max}* (~150 mg/min).

high-capacity SGLT2. In comparison, the higher-affinity and lower-capacity SGLT1 is thought to “clean up” most of the remaining luminal glucose in “later” parts of the proximal tubule (S2/S3 segment) (see Fig. 8.1). In accordance, SGLT2 and SGLT1 have been localized with the use of well-validated antibodies in both rodents and humans to the brush border membrane primarily of the early and late proximal tubule, respectively.^{13–16} In the mouse kidney, the

levels of SGLT1 protein expression in the brush border were highest in S2 segments and somewhat lower in S3 segments in medullary rays and in the outer stripe.¹⁷ In the human kidney, the strongest expression of SGLT1 was found in the S3 segment.¹⁶ In accordance and demonstrating the functional role of SGLT2, free-flow renal micropuncture showed that glucose reabsorption in the early proximal tubule is completely absent in mice lacking SGLT2¹³ (Fig. 8.3). In comparison, fractional glucose reabsorption along the proximal convoluted tubule (PCT) was only slightly reduced from 97% to 94% in mice lacking SGLT1.¹⁸

The phenotype of humans carrying mutations in the genes for SGLT1 (*SLC5A1*) and SGLT2 (*SLC5A2*) demonstrated their distinct quantitative contribution to renal glucose reabsorption. Mutations in SGLT1 cause “Intestinal Glucose Galactose Malabsorption” [Online Mendelian Inheritance in Man (OMIM) 182380] because the active intestinal reabsorption of glucose is mediated by SGLT1.^{19,20} Newborns with mutations in SGLT1 and mice lacking SGLT1¹⁸ can present with life-threatening diarrhea when exposed to dietary galactose or glucose; however, they show little or no glucosuria. In contrast, individuals with mutations in SGLT2 present with persistent “Familial Renal Glucosuria” (OMIM 233100) ranging from 1 to >100 g per day, but they have no intestinal phenotype.²¹ Although mutations in SGLT2 are rare and therefore the consequences are not well studied or fully understood, it is remarkable that no other complications (e.g., urinary infections or impaired kidney function) have been consistently observed in these subjects.^{19,21} This information added to the rationale of developing SGLT2 inhibitors as potentially safe antihyperglycemic drugs (see later). Consistent with the described phenotypes in humans,

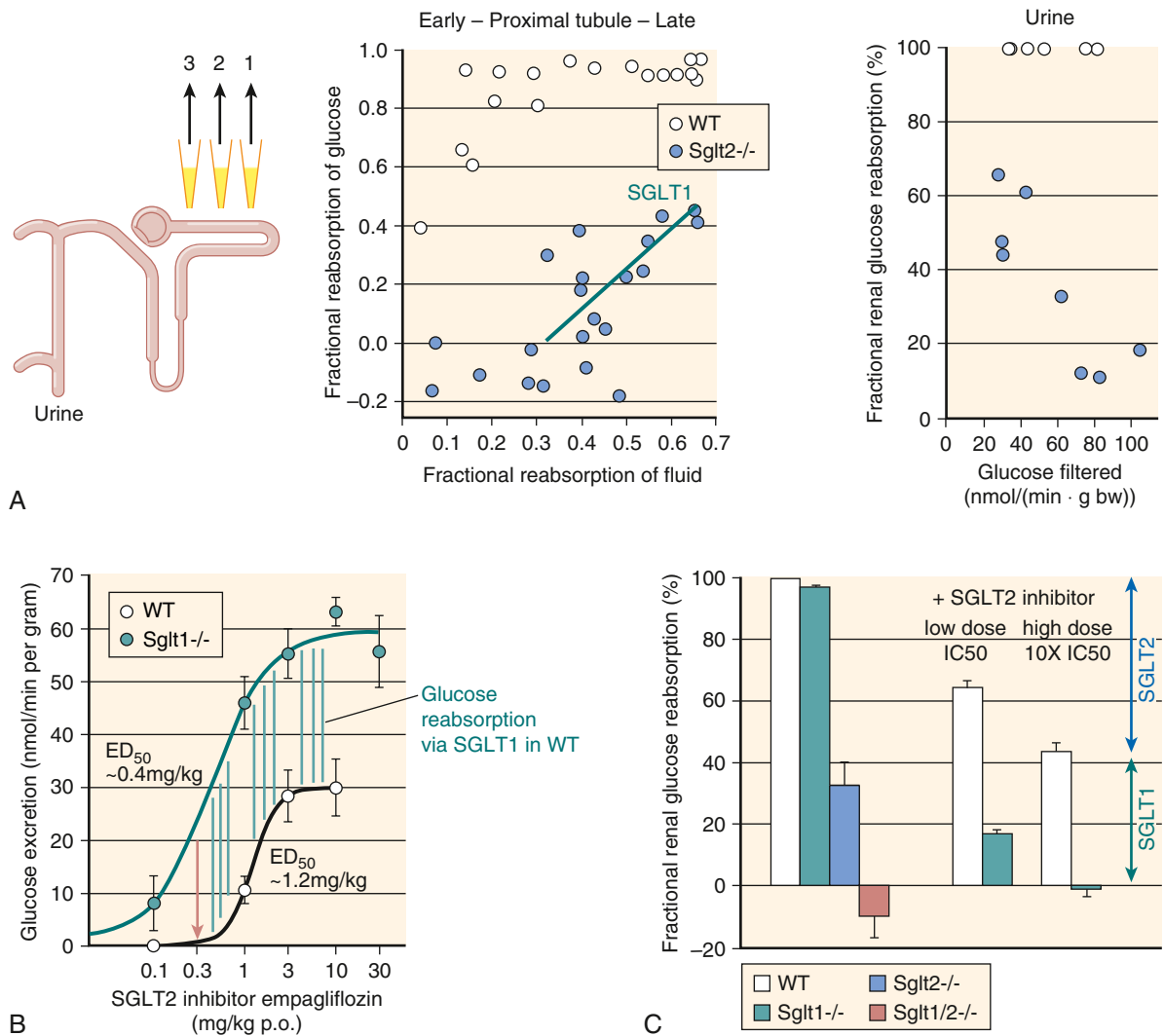


Fig. 8.3 Contribution of SGLT2 and SGLT1 to renal glucose reabsorption. (A) Left panels: Free-flow collections of tubular fluid performed by micropuncture to establish a profile for fractional reabsorption of glucose versus fluid along accessible proximal tubules at the kidney surface. Glucose reabsorption is absent in the early proximal tubule of *Sglt2*^{-/-} mice but enhanced in the later proximal tubule, potentially reflecting compensation by SGLT1-mediated transport. Right panel: Renal inulin clearance studies revealed that the reduction in fractional renal glucose reabsorption in *Sglt2*^{-/-} mice was inversely related to the amount of filtered glucose. (B) In metabolic cage studies, the SGLT2 inhibitor empagliflozin dose dependently increased glucose excretion in wild-type (WT) mice. The response curve was shifted leftward, and the maximum response doubled in *Sglt1*^{-/-} mice. The difference between dose-response curves, which reflects glucose reabsorption via SGLT1 in WT mice, was maintained for higher doses (all vertical lines have same length), consistent with selectivity of the inhibitor for SGLT2 versus SGLT1 in this dose range. Glucosuria is initiated in WT mice when SGLT1-mediated glucose uptake is saturated (red arrow). (C) Renal inulin clearance studies in mice lacking *Sglt1*, *Sglt2*, or both *Sglt1* and *Sglt2* indicated that the glucose reabsorption preserved in *Sglt2*^{-/-} (~40%) is mediated by SGLT1. Application of the SGLT2 inhibitor empagliflozin at low and high doses to establish free plasma concentrations (corresponding to early tubular concentrations) close to IC₅₀ for mouse SGLT2 (~1–2 nM) or 10-fold this concentration confirmed the role of SGLT1 during pharmacologic SGLT2 inhibition. (Data from Vallon V, Platt KA, Cunard R, et al. SGLT2 mediates glucose reabsorption in the early proximal tubule. *J Am Soc Nephrol*. 2011;22:104–112; and Rieg T, Masuda T, Gerasimova M, et al. Increase in SGLT1-mediated transport explains renal glucose reabsorption during genetic and pharmacological SGLT2 inhibition in euglycemia. *Am J Physiol Renal Physiol*. 2014;306:F188–F193.)

genetic and pharmacologic studies in mice showed that under normoglycemic conditions, SGLT2 reabsorbs ~97% of the filtered glucose, whereas SGLT1 mediates the reabsorption of the remaining ~2%–3%^{13,18,22} (see Figs. 8.1 and 8.3).

UNMASKING A SIGNIFICANT GLUCOSE TRANSPORT CAPACITY OF SGLT1 IN THE LATE PROXIMAL TUBULE

In healthy human subjects, similar to the phenotype in rodents, SGLT2 is thought to reabsorb >90% of filtered

glucose, yet they maintain a fractional glucose reabsorption of 40%–50% following application of a selective SGLT2 inhibitor^{23–25} (see Fig. 8.1). This observation is mimicking the phenotype of normoglycemic mice lacking *Sglt2* (*Sglt2*^{-/-}) with mean fractional renal glucose reabsorption of ~40%¹³ (see Fig. 8.3). Follow-up studies demonstrated that the persisting glucose reabsorption is mediated in the downstream late proximal tubule by SGLT1^{13,22} (see Figs. 8.1 and 8.3). These studies demonstrated that SGLT1 provides a significant glucose transport capacity in the late proximal tubule, which, in the normal kidney, remains mostly unused

due to upstream glucose reabsorption by SGLT2. Inhibition of the latter, however, delivers more glucose downstream and unmasks the transport capacity of SGLT1 (see Fig. 8.1). This is also consistent with a high maximal glucose transport rate proposed for human SGLT1 based on in vitro studies.²⁶ As a consequence, combined inhibition of renal SGLT2 and SGLT1 is more glucosuric than inhibition of SGLT2 alone.^{22,27,28} The studies further indicated that SGLT2 and SGLT1 can account for all net glucose reabsorption in the kidney under normoglycemic conditions²² (see Fig. 8.3). Moreover, the data allowed estimating that the basal overall glucose reabsorption capacities for SGLT2 versus SGLT1 in a nondiabetic mouse kidney are in the range of 3:1 to 5:1.²⁹

Clinical Relevance: Inhibition of Renal Glucose Reabsorption as a New Antihyperglycemic Therapy

SGLT2 inhibitors are a class of antihyperglycemic drugs that have been approved in type 2 diabetes mellitus, work independently of insulin, improve glycemic control in all stages of diabetes mellitus with low risk for clinically relevant hypoglycemia, and can be combined with other antidiabetic agents. SGLT2 inhibition lowers glomerular hyperfiltration by a blood glucose-independent mechanism. By inducing a fasting-like metabolic response, reducing glucotoxicity in the early proximal tubule, acting as a diuretic, and lowering blood pressure and glomerular hyperfiltration and hypertension, SGLT2 inhibitors have the potential to induce protective effects on the kidney and cardiovascular system beyond blood glucose control (see Chapters 54 and 41).

MOLECULAR CHARACTERIZATION OF PROXIMAL TUBULAR GLUCOSE TRANSPORT

APICAL GLUCOSE TRANSPORTERS

Crane first proposed in 1960 that active glucose transport in the intestinal epithelium (which expresses SGLT1) was energized by the Na^+ gradient across cell membranes, the so-called Na^+ -glucose cotransport hypothesis (for review, see Wright and colleagues²¹). The Na^+/K^+ -ATPase located on the basolateral membrane is the primary active and ATP-consuming transport step, which lowers cytosolic Na^+ concentrations and establishes the concentration gradient that drives Na^+ uptake, and, secondary, the uptake of other molecules from the luminal surface into proximal tubule cells (Figs. 8.1 and 8.4). This concept was rapidly refined and extended to active transport processes of a diverse range of molecules and ions including Na^+ -glucose cotransport in the kidney.¹⁹

SGLT1 and SGLT2 have been the most intensively studied members of the human SLC5 solute carrier (SLC) family, which now includes 12 members. Six of these are named as SGLTs, varying in their preferences for binding of glucose, galactose, mannose, fructose, myoinositol, choline, short-chain fatty acids, and other anions.¹⁹ All SGLTs have 15 exons, spanning from 8 to 72 kb, which code for 60- to 80-kDa proteins composed of 580 to 718 amino acids.¹⁹ The molecular nature of SGLTs has been largely pioneered by Wright and colleagues (see¹⁹ for review), who identified and cloned SGLT1 and SGLT2, and showed that defects in SGLT1 were associated with intestinal malabsorption of glucose-galactose. Wright's group also defined the

crystal structure of a sodium galactose bacterial isoform in *Vibrio parahaemolyticus* (vSGLT), which allowed better characterization of how Na^+ and sugar transport are coupled: Na^+ binds first to the outside of the transport protein to open the outside gate, thereby permitting outside sugar to bind and be trapped; this is followed by a conformational change and the subsequent opening of the inward gate releases the Na^+ and sugar into the cell cytoplasm. The transport cycle is completed by the change in conformation from an inward-facing ligand-free state to an outward-facing ligand-free state.^{19,30} More recently, the structural basis of human SGLT2 and its inhibition by an SGLT2 inhibitor has been resolved.³¹

The affinity of SGLT1 is similar for glucose and galactose, whereas SGLT2 does not transport galactose or fructose.¹⁹ More recent studies in transfected human embryonic kidney (HEK) 293T cells indicated that the apparent affinities (K_m) for D-glucose are similar for human SGLT1 and human SGLT2, with values of 2 mM and 5 mM, respectively.²⁶ Sugar binding occurs in a Na^+ -dependent manner, and the K_m values for Na^+ transport by human SGLT1 and human SGLT2 are 70 mM and 25 mM, respectively.²⁶ Thus under euglycemic conditions, the glucose concentration in the tubular fluid of the very early proximal tubule (reflecting blood glucose levels) is similar to the K_m of SGLT2, whereas the luminal Na^+ concentration of ~140 mM is much higher than the K_m of SGLT2 and is not rate limiting.

SGLT1 and SGLT2 transport Na^+ and glucose with an Na^+ -glucose coupling ratio of 2:1 and 1:1, respectively.²⁶ The greater Na^+ -glucose coupling ratio of SGLT1 enhances its glucose concentration power and hence the ability of the late proximal tubule to effectively reabsorb glucose despite falling luminal glucose concentrations (see Fig. 8.1). Na^+ -glucose transport is electrogenic, and the membrane potential and driving force are maintained by paracellular Cl^- reabsorption or transcellular K^+ secretion via luminal KCNE1/KCNQ1 channels (see Fig. 8.1).

The mRNA expression of three other members of the SLC5 family that can transport glucose based on in vitro substrate studies has been detected in the kidney.³² SGLT3 (*SLC5A4*) is not a glucose transporter, but glucose can depolarize the plasma membrane in its presence in a saturable, Na^+ -dependent manner, and this effect is inhibited by phlorizin. As such, it has been proposed to be a glucose sensor; however, its expression and function in the kidney remain unclear.³³ SGLT4 (*SLC5A9*) is expressed in the kidney and transports glucose in COS-7 cells but with a lower apparent affinity than mannose (K_i 8 vs. 0.15 mM).³⁴ Thus SGLT4 may primarily be involved in mannose homeostasis. SGLT5 (*SLC5A10*) is a Na^+ -dependent sugar transporter that has a relatively high affinity and capacity for mannose and fructose relative to glucose and galactose.^{35,36} SGLT5 mRNA is highly kidney abundant and expressed in kidney cortex,^{35,37} and studies in knockout mice have indicated that SGLT5 is the major luminal transporter for fructose reabsorption in the kidney.³⁸ For an integrated discussion of kidney glucose and fructose transport and metabolism, see review by Vallon and Nakagawa.³⁹

BASOLATERAL GLUCOSE TRANSPORTERS

In the healthy kidney, the glucose that is being reabsorbed by proximal tubule cells is not linked to appreciable glucose metabolism in these cells. This is due to the fact that most

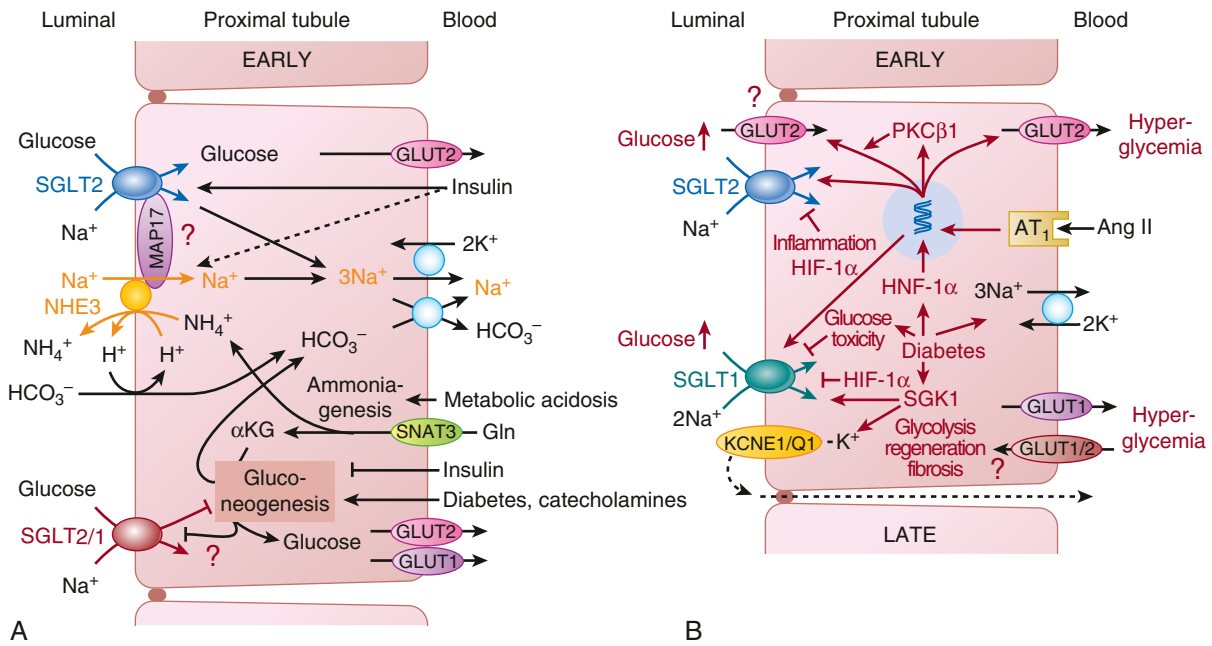


Fig. 8.4 Regulation of glucose transport in the proximal tubule. (A) Insulin is a physiologic stimulator of SGLT2, which may serve to maximize renal glucose reabsorption capacity in situations of increased blood glucose levels (e.g., after a meal). At the same time, enhanced Na⁺-glucose uptake and insulin suppress renal gluconeogenesis. The latter, in contrast, is stimulated by fasting, which may involve increased catecholamine levels. In metabolic acidosis, the increase in gluconeogenesis from glutamine (*Gln*) is linked to the formation of (1) ammonium (NH₄⁺), a renally excreted acid equivalent, and (2) new bicarbonate, which is taken up into the circulation. The Na⁺-H⁺ exchanger NHE3 contributes to apical H⁺/NH₄⁺ secretion and Na⁺/bicarbonate reabsorption. SGLT2 and NHE3 are both stimulated by insulin to enhance Na⁺ and glucose reabsorption, and their functions may be positively linked through the scaffolding protein MAP17. (B) Diabetes increases luminal glucose delivery to both SGLT2- and SGLT1-expressing segments. Glucose transporters GLUT2 and GLUT1 mediate glucose transport across the basolateral membrane, but GLUT2 may also translocate to the apical membrane in diabetes. Angiotensin II (*Ang II*), serum, glucocorticoid-inducible kinase SGK1, hepatocyte nuclear factor HNF-1α, and protein kinase C PKCβ1 promote glucose reabsorption in the diabetic kidney, whereas hypoxia-induced HIF-1α, inflammation, and excessive intracellular glucose levels may be inhibitory. Basolateral glucose uptake via GLUT1/2 may be involved in glycolysis and tubule regeneration after injury as well as hyperglycemia-induced TGF-β.

glucose is reabsorbed in the early PCT (S1 segment), but these cells lack significant capacity for aerobic and anaerobic glycolysis.⁴⁰⁻⁴² Thus glucose that is taken up across the luminal membrane or formed within proximal tubule cells (see later) exits across the basolateral membrane into the interstitium by concentration-driven facilitative glucose transporters, GLUT2 and GLUT1 (see Fig. 8.1) and subsequently enters the peritubular capillaries by convection through fenestrated endothelial cells. GLUT2, the low-affinity (Km; 15–20 mM) “liver” transporter, is primarily expressed in the PCT (S1/S2 segments), but GLUT2 mRNA has also been detected in the proximal straight tubule (PST) (S3 segment).⁴³ GLUT2 is thought to be the dominant transporter involved in basolateral exit of glucose derived from apical glucose uptake or gluconeogenesis in the PCT.⁴⁴⁻⁴⁶ In comparison, GLUT1, the high-affinity (1–2 mM) “erythroid/brain” transporter, is expressed along the entire proximal tubule and has been implicated in transcellular glucose transport, particularly in the S3 segments.⁴⁴⁻⁴⁶ Notably, GLUT1 is also expressed in the basolateral membrane of further distal tubule segments and at higher levels than in S3 segments. This includes expression in the medullary thin and thick ascending limbs (TAL) of the rat kidney and at the highest levels in connecting segments and collecting ducts. In the latter, GLUT1 was expressed at the highest level in intercalated cells and to a lesser extent in principal cells.⁴⁶ These findings indicated a good correlation between the

level of GLUT1 expression and the glycolytic activity of the different nephron segments, indicating that, in particular, the more distal tubule segments are taking up glucose for energy supply via basolateral GLUT1. Some studies have used positron emission tomography and α-methyl-4-[F-18]-fluoro-4-deoxy-d-glucopyranoside to monitor glucose transport in mouse kidneys lacking either SGLT1, SGLT2, or GLUT2. The studies confirmed prominent contributions of SGLT2 and SGLT1 to renal glucose uptake. Moreover, renal glucose reabsorption appeared absent in mice lacking GLUT2, consistent with a more prominent role of GLUT2 versus GLUT1 with regard to basolateral glucose exit of glucose in the proximal tubule⁴⁷ (see Fig. 8.1). This is in line with the renal phenotype of patients with mutations in GLUT2 and GLUT1. Loss-of-function mutations in GLUT2 is the basis of the Fanconi-Bickel syndrome, which includes a renal Fanconi syndrome, a proximal tubulopathy consisting of glycosuria, phosphaturia, aminoaciduria, proteinuria, and hyperuricemia.⁴⁸⁻⁵⁰ The observed proximal tubulopathy may be due to intracellular glucose accumulation and glucotoxicity that occurs when basolateral glucose exit is blocked. In comparison, patients with GLUT1 mutations have primarily neurologic symptoms and no renal phenotype has been documented.^{48,51}

In addition to GLUT1 and GLUT2, some of the other 12 members of the *SLC2* gene family have been found in the kidney and may contribute to glucose transport, but little is

known about their quantitative contribution.⁵² For example, GLUT4 mRNA and immunoreactivity were focally localized in the TALH of the loop of Henle, coexpressed with IGF-I and increased by vasopressin treatment, indicating a potential role in local fuel control.⁴³ GLUT5 is strongly expressed in the apical membrane of the S3 segment in the rat kidney but is proposed to transport primarily fructose.^{43,53} GLUT12 can transport glucose, and an apical localization has been reported in distal tubules and collecting ducts, but its quantitative role remains to be determined.⁵⁴

RENAL FORMATION OF GLUCOSE

The kidneys reabsorb the filtered glucose and use glucose as an energy source, but they also generate new glucose. Gluconeogenesis involves the formation of glucose-6-phosphate from precursors such as lactate, glutamine, alanine, and glycerol with its subsequent hydrolysis by glucose-6-phosphatase to generate free glucose that can exit the cell. The healthy human kidneys produce ~15–55 g glucose per day by gluconeogenesis, similar to the liver in the postabsorptive state (i.e., 12–16 hours after the last meal).⁵⁵ Renal gluconeogenesis in the postabsorptive state is stimulated by epinephrine and falling levels of insulin⁵⁵ (see Fig. 8.4). Insulin-induced suppression of gluconeogenic gene expression in the proximal tubule was accompanied by phosphorylation and inactivation of forkhead box transcription factor 1 (FoxO1).⁵⁶ In contrast to the liver, renal gluconeogenesis is probably insensitive to glucagon.⁵⁵ Studies in humans indicated that in the postabsorptive state, renal gluconeogenesis primarily uses lactate as substrate, followed by glutamine, glycerol, and alanine.⁵⁷

In contrast to the uniform stimulation of gluconeogenesis along the entire proximal tubule by starvation, metabolic acidosis enhances gluconeogenesis primarily in S1 and S2 segments.^{41,58,59} Furthermore, gluconeogenesis in response to metabolic acidosis primarily uses glutamine as substrate. During the process of renal gluconeogenesis from glutamine, the conversion of glutamine to glutamate and α -ketoglutarate produces ammonium (NH_4^+), which is excreted into the urine as an acid equivalent. The subsequent pathway from α -ketoglutarate to glucose forms new bicarbonate, which is returned as buffer to the systemic circulation (see Fig. 8.4). The link between proximal tubular ammonium, bicarbonate, and glucose formation explains why acidosis is a prominent stimulus for renal gluconeogenesis.^{55,57}

Apical glucose uptake via SGLT1 or SGLT2 can have an inhibitory influence on the expression of renal gluconeogenic genes (see Fig. 8.4). This effect may serve to prevent glucose overload in the cells of the proximal tubule and has been proposed to involve glucose-induced and sirtuin 1-mediated deacetylation of peroxisome proliferator-activated receptor gamma coactivator 1- α , a coactivator of FoxO1.⁵⁶

In general, it is expected that cytosolic glucose in proximal tubule S3 segments is used for metabolism or leaves the cell via basolateral GLUT1/GLUT2. It has also been hypothesized that glucose generated from lactate in the medullary S3 segment forms part of an intrarenal Cori cycle;⁶⁰ glucose enters the lumen by reversed transport through SGLT1 and is taken up by downstream tubular segments, where it is used as energy substrate for glycolysis (e.g., in medullary TAL),

and the formed lactate is being returned to the neighboring S3 segment as a substrate for gluconeogenesis. Studies in human proximal tubule segments indicated that in contrast to S1 segments, lactate appears to be a better gluconeogenic precursor than glutamine in S2 and S3 segments.⁶¹ Moreover, studies in mice, rats, and humans indicated that the luminal membrane of the thick ascending limb (and the adjacent macula densa) express SGLT1.^{15,17,62} Further studies are needed to determine the potential role of SGLT1 in these structures including a proposed Cori cycle.

Taken together and under normal conditions, the PCT is the site of greatest renal glucose reabsorption and generation. The ability of early proximal tubular segments for gluconeogenesis but inability to metabolize glucose prevents a futile cycle. The reabsorption of filtered glucose and renal gluconeogenesis provide an energy source to distal tubular segments, primarily in the renal medulla, and returning the glucose to the systemic circulation helps to maintain blood glucose levels, particularly in the postabsorptive state. In addition, renal gluconeogenesis is closely linked to the renal response to metabolic acidosis. See Chapter 5 for additional discussion of the role of gluconeogenesis in renal metabolism.

RENAL GLUCOSE TRANSPORT IN DISEASE STATES

GLUCOSE TRANSPORT IS INCREASED IN THE DIABETIC KIDNEY

Therapies for type 2 diabetes mellitus (T2DM) have included drugs that target the liver, small intestine, adipose tissue, skeletal muscle, and/or pancreas. Many of these therapies, including insulin, have difficulties to establish adequate glycemic control without the potential for relevant unwanted side effects like hypoglycemia and weight gain and may not reduce cardiovascular complications.⁶³ The following sections outline how glucose transport changes in the diabetic kidney, its implications for diabetic kidney function, and how targeting renal glucose transport is an antihyperglycemic and kidney-protective therapy.

Diabetes mellitus is associated with increased blood glucose levels. This enhances the amounts of glucose filtered by the kidneys. The early phase of diabetes is often associated with an increase in GFR or glomerular hyperfiltration (see later), which further increases the tubular glucose load. At the same time, the tubular capacity to reabsorb glucose increases by ~20%–30% to ~500–600 g/day in patients with T2DM^{2,64} and type 1 diabetes (T1DM).¹ Moreover and despite increased blood glucose levels, diabetes also enhances renal gluconeogenesis.⁵⁵ The latter can be the consequence of diabetes-associated metabolic acidosis, and the induced gluconeogenesis involves metabolism of glutamine to glucose associated with the generation of ammonia and bicarbonate⁵⁵ (see Fig. 8.4). Other potential triggers for gluconeogenesis in diabetes include the activation of the sympathetic nervous system, the reduced insulin levels observed in T1DM, enhanced circulating fatty acids, or cells that are glucose “starved” due to insulin resistance (T2DM).

Increasing renal glucose reabsorption in response to a rise in filtered glucose makes sense with regard to energy substrate conservation. Moreover, the further distal segments may need/use more glucose as an energy

substrate to reabsorb the load of salt and other compounds, which is increased due to glomerular hyperfiltration. Renal glucose retention and enhanced glucose formation become maladaptive in diabetes, however, when they sustain hyperglycemia (see Fig. 8.4). When blood glucose levels increase to the point that the filtered load exceeds the T_m or tubular transport capacity for glucose, then the surplus is excreted in the urine. In this regard, the kidney provides a safety valve that can prevent extreme hyperglycemia. The safety valve, however, only opens at rather high blood glucose levels (>15 mmol/L) (see Fig. 8.2) and only works as long as glomerular filtration is maintained. SGLT2 inhibitors cause this valve to open at lower blood glucose levels.

GLUCOSE TRANSPORTERS IN THE DIABETIC KIDNEY

The levels of protein expression and activity of SGLT2, SGLT1, GLUT2, and potentially GLUT1 determine the capacity of renal glucose reabsorption, and their upregulation may explain the increased glucose transport maximum that can be observed in diabetes. The available preclinical and human studies reported increased, unchanged, or reduced renal glucose transporter expression and/or activity in diabetes or under high-glucose conditions.⁶⁵ The observed different responses may reflect different diabetes models, metabolic states, levels of kidney injury, other factors that regulate the expression of these transporters, the use of nonselective antibodies, or dissociations between mRNA and protein expression.

SGLT2 AND GLUT2

Using knockout mice as gold standard negative antibody controls, the renal protein expression of SGLT2 was found to be increased by 40% to 80% in the early hyperglycemic stages of genetic mouse models of T2DM (db/db) and T1DM (Akita).^{66,67}

Consistent with a potential concerted regulation of luminal and basolateral glucose transport, upregulation of GLUT2 expression has been reported in renal proximal tubules in diabetic rats.⁶⁸⁻⁷¹ Notably, studies in streptozotocin (STZ)-induced T1DM in rats proposed targeting of GLUT2 (but not GLUT1) also to the brush border membrane of proximal tubules.^{72,73} The latter may be linked to protein kinase C $\text{PKC}\beta 1$ activation⁷³⁻⁷⁵ and may implicate facilitative glucose transport, together with SGLT2 and SGLT1 in the increased glucose reabsorption across the apical membrane of proximal tubules in the diabetic kidney (see Fig. 8.4). On the other hand, renal clearance studies in genetic models of T1DM and T2DM in mice found that coinhibition of SGLT2 and SGLT1 prevented any net glucose reabsorption in the kidney.⁷⁶

The available data on changes in glucose transporters in diabetic patients are sparse and also variable. Primary cultures of human exfoliated proximal tubular epithelial cells harvested from fresh urine of patients with T2DM showed an increased glucose uptake associated with increased protein expression of SGLT2 and GLUT2.⁷⁷ An increase in SGLT2 protein expression has also been reported in fresh kidney biopsies of patients with T2DM and advanced nephropathy.⁷⁸ On the other hand, the mRNA expression of SGLT2 and GLUT2 was slightly lower in 19 patients with T2DM and preserved kidney function as compared with 20 nondiabetic patients matched for age and estimated GFR (eGFR), all being subjected to nephrectomy.⁷⁹ Similar

results were reported for SGLT2 and GLUT2 mRNA in another set of patients with T2DM, but the results did not reach statistical significance.⁸⁰

If an increase in SGLT2 expression occurs in the diabetic kidney, then it may simply reflect overall growth and hypertrophy of the diabetic proximal tubule and the associated increase in transport machinery,^{81,82} and this may be exaggerated on the single-nephron level with advanced nephropathy when nephrons are lost and the remaining nephrons aim to compensate. Moreover, hyperinsulinemia associated with extrarenal insulin resistance in obesity and T2DM (as well as postprandial hyperinsulinemia) may phosphorylate SGLT2 at Ser624⁸³ and contribute to enhanced renal SGLT2 activity⁸⁴ (see Fig. 8.4). Furthermore, upregulation of SGLT2 expression in diabetic rats has been linked to activation of Ang II AT1 receptors⁸⁵ and the transcription factor, hepatocyte nuclear factor HNF-1 α .⁸⁶ The latter, as well as HNF-3 β , have also been implicated in renal GLUT2 upregulation⁶⁹ (see Fig. 8.4). Notably, pharmacologic inhibition of SGLT2 in normoglycemic mice also increased renal membrane SGLT2 protein expression,⁶⁷ possibly reflecting negative feedback regulation by intracellular glucose levels. Along this line, if renal SGLT2 expression is reduced in the diabetic kidney, this may be due to enhanced diabetes-induced proximal tubular gluconeogenesis (see Fig. 8.4) or reflect more severe tubular hypoxia or damage.⁸⁷⁻⁸⁹

SGLT1 AND GLUT1

The renal expression of SGLT1 protein appears to vary among genetic mouse models of diabetes: Renal SGLT1 protein expression was found to be increased in leptin-deficient ob/ob mice,⁹⁰ a model of T2DM, and reduced in Akita mice, a model of T1DM;⁶⁶ the latter study used knockout mice as negative antibody control. In contrast to SGLT2 (see earlier), insulin stimulation slightly decreased SGLT1-mediated Na^+ -glucose transport in HEK-293T cells,⁸³ indicating differences in the regulation of these two transporters. In contrast to the strong increase in SGLT2 (see earlier), SGLT1 protein was not significantly changed in fresh kidney biopsies of patients with T2DM and nephropathy in comparison with nondiabetic controls.⁷⁸ The interpretation of renal SGLT1 mRNA expression data may be complicated by the observation that mRNA and protein expression can dissociate, at least in mouse kidneys.⁹¹

GLUT1 protein expression was downregulated in proximal tubules isolated from rat cortices at 2 and 4 weeks after STZ⁷¹ but increased in kidneys of rats at 30 weeks after STZ.⁵⁴ A study in patients with T2DM and preserved kidney function reported that in whole renal tissue, GLUT1 mRNA expression was slightly lower as compared with nondiabetic patients.⁷⁹

Why should diabetes reduce renal SGLT1 expression? Although this would permit the renal glucose “valve” to open earlier (and make SGLT2 inhibitors more efficacious, see later), this may not be the kidneys’ intention. A reduced renal SGLT1 protein expression was also observed in response to genetic or pharmacologic SGLT2 inhibition in nondiabetic mice.^{13,67} These conditions and diabetes share an enhanced glucose load to the late proximal tubule. In vitro studies in proximal tubule cells indicated that high glucose can reduce SGLT expression and Na^+ -glucose cotransport

activity via enhanced oxidative stress.⁹² Studies in a model of pig epithelial tubular cells (LLC-PK1) showed that hypoxia can diminish SGLT1 (and SGLT2) protein expression by activation of hypoxia-inducible factor-1 α (HIF-1 α).⁸⁸ Thus an increased glucose load to the outer medullary S3 segment enhances Na⁺-glucose reabsorption and thereby hypoxia, which may downregulate SGLT1 to limit oxygen-consuming transport work and glucotoxicity in this segment, which has a high sensitivity to acute injury⁸⁷ (see Fig. 8.4).

In comparison, an increase in SGLT1 expression in the diabetic kidney would further increase the renal glucose reabsorption capacity but may put the S3 segment at risk of hypoxia and enhanced glucotoxicity. Studies in Akita diabetic mice indicated that the serum and glucocorticoid-inducible kinase SGK1 may stimulate SGLT1 activity and glucose reabsorption in PSTs.⁹³ SGK1 could also promote proximal tubular glucose reabsorption by enhancing the activity of luminal K⁺ channels (KCNE1/KCNQ1), which maintain the electrical driving force during electrogenic Na⁺-glucose cotransport^{94,96} (see Fig. 8.4). SGK1 was upregulated in proximal tubules in patients with diabetic nephropathy.⁹⁷

Transport functions in proximal tubules require high turnover of ATP, which, under normal conditions, is derived primarily through mitochondrial oxidative phosphorylation.^{41,98} This may change in pathophysiologic situations with impaired mitochondrial function, when glycolysis may be enhanced and contribute to maintaining ATP. For example, a shift to glycolysis has been proposed in proximal tubules regenerating from acute kidney injury (AKI), as well as proximal tubules undergoing atrophy.⁹⁹ This metabolic switch to glycolysis occurred early during proximal tubule regeneration and was reversed during successful tubular recovery, but it persisted and became progressively more severe in tubule cells that failed to redifferentiate. Tubular upregulation of HIF-1 α in mice enhanced renal GLUT1 mRNA expression; this was associated with less oxygen consumption and increased glycolysis.¹⁰⁰ Thus hypoxia may increase basolateral GLUT1-mediated facilitative uptake of glucose, which is then used for glycolysis and recovery. Hypoxia-induced GLUT1 likely applies to distal tubule segments but may also be relevant for medullary S3 segments.⁹⁹ In this regard, studies in the proximal tubular cell line LLC-PK1, cultured and polarized on porous tissue culture inserts, showed that basolateral exposure to 25 mmol/L D-glucose enhanced glucose uptake via GLUT1 and the subsequent intracellular metabolism of glucose enhanced TGF- β 1 synthesis and secretion; this was not observed in response to apical glucose exposure.¹⁰¹ These in vitro studies suggest that it may be the hyperglycemia-induced persistent uptake of glucose via basolateral GLUT1 (or GLUT2), rather than the filtered glucose, that affects the tubular synthesis of TGF- β 1 and thereby the development of tubulointerstitial fibrosis and tubular growth (see Fig. 8.4).

INHIBITION OF RENAL GLUCOSE REABSORPTION AS A ANTIHYPERGLYCEMIC THERAPY

When blood glucose levels increase to the point that the filtered load exceeds the transport capacity of the tubular system, then the surplus is excreted in the urine. This renal

“safety valve” can prevent extreme hyperglycemia. When SGLT2 is inhibited, the reabsorptive capacity for glucose declines to the residual capacity of SGLT1, which equals ~80 g/day. In other words, SGLT2 inhibition causes the renal safety valve to open at a lower threshold (see Fig. 8.2), maintaining glucose levels in the euglycemic and moderately hyperglycemic range. Several SGLT2 inhibitors are approved as glucose-lowering agents for subjects with T2DM. The following sections discuss the role of SGLT2 in the pathophysiology of renal glucose reabsorption and outline the unexpected logic of inhibiting SGLT2 in the diabetic kidney.^{81,102,103} This includes the counterproductive enhancement of renal glucose reabsorption via SGLT2 in diabetes, as well as a brief discussion of the basic mechanisms that link primary inhibition of Na⁺-glucose cotransport in the kidney to secondary beneficial consequences on metabolism, the kidneys, and the cardiovascular system.

Long-term access to excessive exogenous energy resources is not part of human evolution. As a consequence, it may not come as a surprise that the body's responses to excess exogenous energy resources can be maladaptive. In contrast, the body's responses to environments with scarce energy resources have been intensively tested and refined during evolution for the survival of the organism. Therefore targeting metabolism in the “periphery” by inhibiting renal glucose reabsorption and spilling glucose as an energy resource into the urine, which induces a fasting-like state and activates proven counterregulatory mechanisms to readjust metabolism, provides unique benefits as an antihyperglycemic approach.⁸¹ This is supported by clinical outcome studies using an SGLT2 inhibitor on top of standard of care in patients with high cardiovascular risk that demonstrated protective effects with regard to CKD and heart failure in patients with type 2 diabetes and, more recently, also in nondiabetic patients.¹⁰⁴⁻¹⁰⁸

Phlorizin is a competitive inhibitor of SGLT2 and SGLT1, the former with a 10-fold higher affinity.^{19,26,367} SGLT1 is the primary pathway for glucose reabsorption in the intestine,¹¹⁰ and as a consequence, oral administration of phlorizin causes diarrhea. In comparison, SGLT2 is expressed primarily in kidney proximal tubule.^{16,91} Therefore inhibition of renal glucose transport became practical when phlorizin derivatives were developed that are specific for SGLT2, have good oral bioavailability, and are suitable for once-daily dosing.¹¹¹

SGLT2 inhibitors act on their target from the extracellular surface of the cell membrane¹¹² and reach their target by glomerular filtration and, as indicated for empagliflozin, also by tubular secretion.¹¹³ SGLT2 inhibitors induce a sustained urinary glucose loss of 40 to 80 g/day.^{111,114,115} In patients with T2DM, this is associated with a decrease in HbA1C levels by 0.5% to 0.7% at 12 weeks of treatment, and this effect persisted for up to 52 weeks.¹¹⁶ The higher the blood glucose level and GFR, the more glucose is filtered and reabsorbed and, as a consequence, can be excreted in response to SGLT2 blockade. Thus SGLT2 blockers naturally have a greater efficacy when it is desirable for them to be more efficacious.^{13,117,118} By lowering blood glucose levels and body weight, SGLT2 inhibitors improve β -cell function and sensitivity to insulin in patients and rodent models with T2DM.^{117,119-122} Because the renal mechanism of action of SGLT2 inhibitors is independent of insulin, their efficacy does not decline with progressive β -cell dysfunction and/

or insulin resistance and SGLT2 inhibitors act synergistically with other blood glucose-lowering agents.¹¹¹ The main side effect of SGLT2 inhibitors is an increased risk of genitourinary infections due to the glucosuric effect.¹²³

HOW CAN INHIBITION OF RENAL GLUCOSE TRANSPORT PROTECT THE KIDNEY AND THE CARDIOVASCULAR SYSTEM?

By reducing hyperglycemia, SGLT2 inhibitors have the potential to reduce glucotoxicity in the kidney and extrarenal organs.^{124,125} In accordance, studies in diabetic rodent models with severe hyperglycemia have shown that SGLT2 inhibition can reduce growth, lipid accumulation, inflammation, and

injury of the diabetic kidney and improve autophagy secondary to strong blood glucose-lowering effect^{66,67,78,90,126-129} (Fig. 8.5). The observed smaller effect of SGLT2 inhibitors on blood glucose in the clinical trials contributes to, but cannot fully explain, the profound and rapid impact on cardiac and kidney outcomes observed in nondiabetic patients and appears insufficient to fully explain the rapid beneficial effect on heart failure detectable within a few months.

SGLT2 INHIBITION LOWERS BODY WEIGHT AND HAS A LOW HYPOGLYCEMIA RISK

In patients with T2DM, the glucosuric effect of SGLT2 inhibition is associated with a 2- to 3-kg lower body weight.

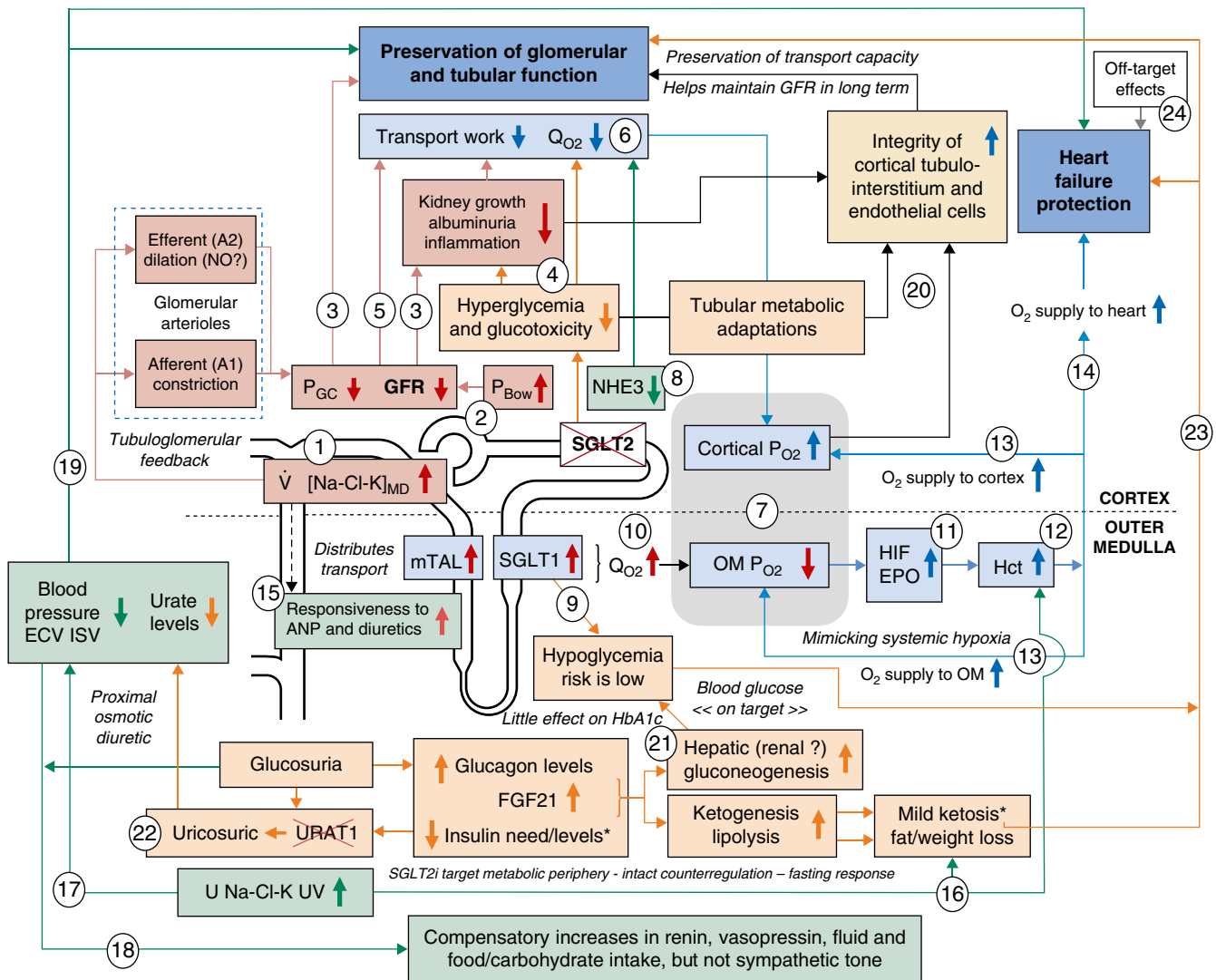


Fig. 8.5 Kidney-protective mechanisms of SGLT2 inhibition. SGLT2 inhibition reduces the reabsorption of glucose and Na^+ in the early proximal tubule. This increases the delivery of NaCl and K ($[\text{Na-Cl-K}]_{\text{MD}}$) and fluid (V) to the macula densa, which lowers glomerular filtration rate (GFR) and glomerular capillary pressure (P_{GC}) through tubuloglomerular feedback (TGF) and by increasing hydrostatic pressure in the Bowman space (P_{Bow}). Lowering P_{GC} & GFR and hyperglycemia protects glomerular and tubular function by reducing filtration of albumin, decreased tubular growth, inflammation, and reduced tubular transport work, which lowers cortical oxygen demand. Tubular transport work and toxicity are also reduced by inhibition of Na-H-exchanger NHE3. Shifting glucose and Na^+ reabsorption to SGLT1 and medullary thick ascending limb (mTAL) lowers oxygen level in the outer medulla, which may activate hypoxia-inducible factor (HIF) and enhance erythropoietin release. The resulting increase in hematocrit (Hct)(12) improves O_2 delivery to kidney medulla and cortex. More delivery of NaCl and fluid downstream of early proximal tubule may facilitate responsiveness to atrial natriuretic peptide (ANP) and enhance the diuretic effect of SGLT2 inhibition, leading to reduced blood pressure. SGLT2 inhibitors are uricosuric, which has the potential to further protect the kidney and heart. (This figure was modified with permission from Vallon V. Glucose transporters in the kidney in health and disease. *Pflügers Arch.* 2020;472[9]:1345–1370.)

Although the diuretic effect and fluid loss may contribute to the initial weight loss, the majority of the steady-state weight loss with SGLT2 inhibitor treatment is due to fat loss, including visceral and subcutaneous fat¹³⁰⁻¹³² (see Fig. 8.5), as a consequence of a shift in substrate utilization from carbohydrates to lipids.^{117,133,134} The released free fatty acids are used by the liver to form ketone bodies and thus increase ketogenesis.¹³⁵ SGLT2 inhibitors may improve cardiac and kidney outcomes in part by increasing plasma levels of ketone bodies like β -hydroxybutyrate, which are used as additional energy substrates to improve the performance of cardiac myocytes (or kidney cells) in diabetes mellitus^{136,137} (see Fig. 8.5). Importantly, SGLT2 inhibitors can increase the risk of diabetic ketoacidosis,¹³⁵ particularly when the drugs are used off label in patients with T1DM.¹³⁵

In contrast to insulin therapy, SGLT2 inhibitors do not increase the incidence of hypoglycemia.^{104-106,116} This is because they become ineffective at lowering blood glucose any further once the filtered glucose load falls to ~80 g/day, which can be handled by renal SGLT1 (see Fig. 8.1 and 8.5). In addition, SGLT2 inhibitors leave the metabolic counterregulation intact, which can increase plasma glucagon concentrations and subsequently endogenous hepatic but also kidney glucose production (gluconeogenesis) in patients with T2DM.^{117,120,138} This is potentially relevant for cardiovascular outcomes because episodes of hypoglycemia can reduce the cardioprotective effects of antihyperglycemic therapy.¹³⁹

SGLT2 INHIBITION LOWERS BLOOD PRESSURE AND IMPROVES HYPERURICEMIA

A meta-analysis of patients with T2DM treated with SGLT2 inhibitors found a consistent decrease in systolic blood pressure of 3 to 6 mm Hg.¹¹⁶ The magnitude of this blood pressure effect is expected to have cardiovascular protective consequences, particularly in high-risk patients.¹⁴⁰ The blood pressure-lowering effect of SGLT2 inhibition relates to the reduction in body weight and a modest glucose-based osmotic diuresis (100–470 mL/day) and a small natriuretic effect.^{124,141-144} The lower blood pressure and an associated modest reduction in plasma volume¹⁴⁵ may rapidly reduce cardiac preload and afterload and thereby contribute to the rapid beneficial effects in heart failure patients¹⁰⁴ (see Fig. 8.5). SGLT2 is functionally linked to the Na-H-exchanger NHE3,^{138,146-148,149} such that SGLT2 inhibition induces natriuresis and lowers blood pressure in part by inhibiting NHE3 in the proximal tubule (see Figs. 8.4 and 8.5).

Beneficial renal and cardiovascular effects of SGLT2 inhibition may also be due to a plasma uric acid-lowering effect.¹⁵⁰ The uricosuric effect of SGLT2 inhibitors is positively related to the increase in tubular and urinary glucose delivery, as observed in healthy subjects and patients with T2DM^{151,152} (see Fig. 8.5) and may involve an interaction with the proximal tubule luminal urate transporter URAT1.^{152,153}

With regard to cardiac protection, more evidence is accumulating that supports the concept that SGLT2 inhibitors may also work through off-target effects on cardiac transporters like NHE-1, Nav1.5, or SGLT1¹⁵⁴ (see Fig. 8.5), but much needs to be learned.¹⁰³

SGLT2 INHIBITION LOWERS GFR INITIALLY TO PRESERVE IT LONG TERM

Glomerular hyperfiltration, which is observed in a subset of patients at the onset of T1DM and T2DM, can increase the risk of developing diabetic nephropathy later.¹⁵⁵ Glomerular hyperfiltration increases transport work and oxygen consumption in the diabetic kidney, and lowering GFR has opposite effects.¹⁴¹

According to the “tubular hypothesis,” glomerular hyperfiltration in diabetes is explained by primary tubular hyperreabsorption (for review, see Vallon and Thomson¹⁵⁶). Moderate levels of hyperglycemia increase proximal tubular reabsorption by providing more substrate for Na⁺-glucose cotransport via SGLT2 and SGLT1 and by causing the tubule to grow, which enhances the transport machinery and capacity. The increased reabsorption reduces the NaCl and fluid delivery to the downstream macula densa, which senses this reduction and causes GFR to increase through the normal physiologic action of tubuloglomerular feedback (TGF) (Fig. 8.6). The primary role of the TGF is to stabilize the NaCl and fluid delivery downstream of the macula densa and thereby facilitate the fine regulation of NaCl and fluid balance in the distal nephron by neurohumoral control. A secondary consequence of this TGF physiology is that the mechanism contributes to the autoregulation of GFR and renal blood flow. Moreover, it makes GFR responsive to primary changes in tubular transport upstream of the macula densa, like in the diabetic kidney. A primary increase in proximal reabsorption also reduces the distal tubular flow rate, which increases GFR by lowering tubular back pressure (i.e., the hydrostatic pressure in Bowman space) and thereby increasing the effective glomerular filtration pressure (see Fig. 8.6). Mathematic modeling indicates that TGF and the changes in tubular back pressure contribute equally to the increase in GFR in diabetes.¹⁵⁷

Conversely, SGLT2 inhibition attenuates proximal tubule hyperreabsorption in the diabetic kidney and thereby lowers diabetic glomerular hyperfiltration (see Figs. 8.5 and 8.6). This has been shown in micropuncture studies in rats using direct application of phlorizin into the Bowman space¹⁵⁸ and by acute or chronic systemic application of selective SGLT2 inhibitors.¹⁵⁹ In accordance, pharmacologic or genetic inhibition of SGLT2 suppressed hyperfiltration on the whole-kidney level in diabetic mice.^{66,67} Consistent with the proposed local mechanism, the suppression of diabetic hyperfiltration in response to SGLT2 inhibition was associated with an increase in the NaCl concentration at the macula densa^{158,159} and in the hydrostatic pressure in the Bowman space,¹⁵⁸ and it was independent of effects on blood glucose^{66,158,159} (see Fig. 8.5).

When macula densa cells sense an increase in luminal NaCl, the resulting TGF-induced ATP release promotes local formation of adenosine, which primarily constricts the afferent arteriole via adenosine A₁ receptors but can also dilate the efferent arteriole via adenosine A₂ receptors (see Fig. 8.5). Both effects are expected to lower glomerular capillary pressure (P_{GC}). Micropuncture of glomerular capillaries in rats with T1DM established that the SGLT2 inhibitor ipragliflozin indeed reduced GFR and P_{GC}.¹⁶⁰ However, the changes in P_{GC} and GFR were not strictly correlated, consistent with the asymmetry of afferent and efferent arteriolar TGF responses and their consequences on

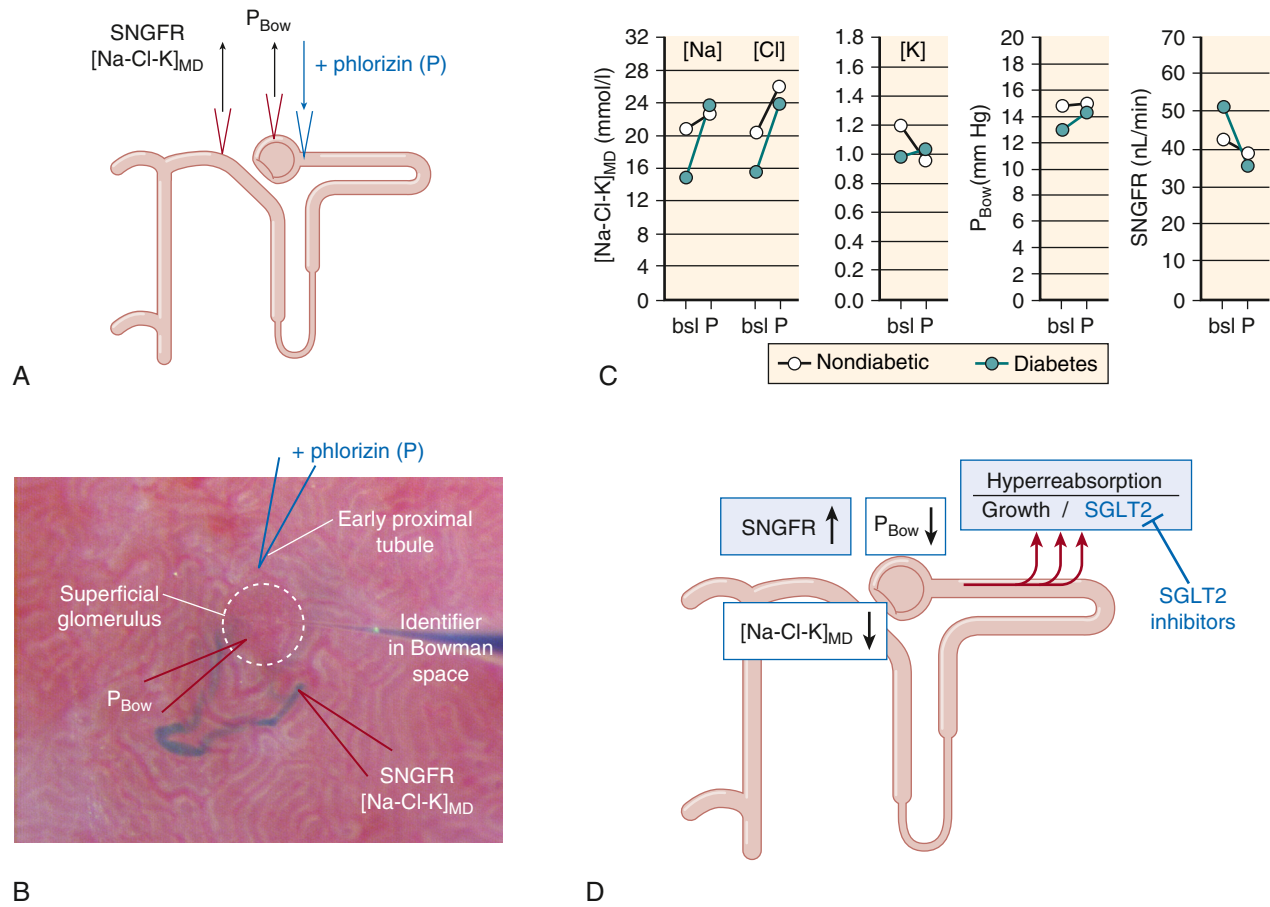


Fig. 8.6 The tubular hypothesis of diabetic glomerular hyperfiltration: effect of SGLT2 inhibition. (A, B) In vivo micropuncture studies in rats with superficial glomeruli were performed in nondiabetic and streptozotocin-diabetic rats.¹⁵⁸ Small amounts of blue dye were injected into the Bowman space to determine nephron configuration, including the first proximal tubular loop and the early distal tubule close to the macula densa. Tubular fluid was collected close to the macula densa to determine 1. the tubuloglomerular feedback signal ($[Na-Cl-K]_{MD}$) and 2. single-nephron glomerular filtration rate (SNGFR) by inulin clearance. The Bowman space was punctured to determine the hydrostatic pressure (P_{Bow}). Measurements were performed under control conditions and following application of the SGLT2/SGLT1 inhibitor phlorizin into the early proximal tubule (i.e., without changing systemic blood glucose levels). (C) Basal measurements (bsl) revealed that glomerular hyperfiltration in diabetes was associated with reductions in $[Na-Cl-K]_{MD}$ and P_{Bow} . Adding phlorizin (P) had a small effect in nondiabetic rats but normalized $[Na-Cl-K]_{MD}$, P_{Bow} , and SNGFR in diabetes. (D) Diabetes induces a primary hyperreabsorption in the proximal tubule due to tubular growth and enhanced Na^+ -glucose cotransport, which, through tubuloglomerular feedback ($[Na-Cl-K]_{MD}$) and reducing tubular back pressure (P_{Bow}), causes glomerular hyperfiltration. SGLT2 contributes to tubular hyperreabsorption, and as a consequence, SGLT2 inhibition mitigates these changes and lowers glomerular hyperfiltration. (This figure was modified with permission from Vallon V, Thomson SC. Targeting renal glucose reabsorption to treat hyperglycaemia: the pleiotropic effects of SGLT2 inhibition. *Diabetologia* 2017;60:215–225.)

GFR and P_{GC} .¹⁶⁰ Thus SGLT2 inhibition can induce a robust reduction in P_{GC} even when GFR decreases only slightly and vice versa. This may have implications in advanced CKD ($GFR < 30 \text{ mL}/[\text{min} \times 1.73 \text{ m}^2]$), where the initial GFR drop in response to SGLT2 inhibition can be smaller, but the kidney-protective effect is preserved, possibly due to a robust effect on the efferent arteriole and predominant reduction in P_{GC} .¹⁶⁰

The initial GFR-lowering effect has been confirmed in humans. The SGLT2 inhibitor empagliflozin decreased GFR by 19% in T1DM patients with baseline hyperfiltration independently of lowering blood glucose levels.¹¹⁸ The SGLT2 inhibitor canagliflozin initially lowered eGFR in patients with T2DM and basal eGFR of $\geq 55 \text{ mL}/\text{min}/1.73 \text{ m}^2$. Following this initial dip, eGFR remained stable over the following weeks and months in the canagliflozin-treated group such that eGFR was better preserved after 2 years of follow-up and associated with reduced urinary

albumin-to-creatinine ratios than in the control group, which had been treated with glimepiride to achieve similar blood glucose control.¹⁶¹ Similar biphasic eGFR profiles (i.e., initial dip followed by subsequent better preservation) have been shown for other SGLT2 inhibitors.⁷⁶

Surviving nephrons in advanced stages of CKD are assumed to hyperfilter as a way of compensation for the reduced nephron number and thus maintain a high-glucose load on the level of the single nephron. This should preserve the acute GFR-lowering effect of SGLT2 inhibition, even if the effect on overall glucose homeostasis is attenuated. Importantly, lowering single-nephron glomerular hyperfiltration in CKD and thereby the oxygen-consuming transport work may help to preserve the integrity of the remaining nephrons and overall kidney function over the long term (see Fig. 8.5). The additive effect of inhibitors of the renin-angiotensin system and of SGLT2 inhibitors is consistent with the concept that angiotensin II blockade

primarily dilates the efferent arteriole, whereas SGLT2 inhibition primarily constricts the afferent arteriole.

It has been discovered that the macula densa senses an increased glucose delivery via luminal SGLT1, which then activates nitric oxide synthase 1 (NOS1), and the resulting increase in NO formation blunts the afferent arteriolar vasoconstrictor effect of TGF and thereby contributes to diabetic hyperfiltration.^{62,162} Thus the increase in macula densa glucose delivery in response to an SGLT2 inhibitor can activate the macula densa-SGLT1-NOS1 pathway¹⁶² and, thereby, potentially limit the initial fall in GFR. On the other hand and considering the close proximity of the macula densa to both afferent and efferent arterioles, this pathway may also dilate the efferent arteriole, potentially in settings with endothelial dysfunction and low efferent NO tone.

SGLT2 INHIBITORS INDUCE TUBULAR METABOLIC ADAPTATIONS

In patients with T2DM and albuminuria, dapagliflozin increased urinary metabolites linked to mitochondrial metabolism, potentially indicating that dapagliflozin improves mitochondrial function in the diabetic kidney.¹⁶³ Consistent with this, scRNA sequencing of proximal tubules in db/db mice indicated that while RAS blockade is more antiinflammatory/antifibrotic, SGLT2 inhibition affected genes related to mitochondrial function.¹⁶⁴ ScRNA sequencing from research kidney biopsies from young persons with T2DM indicated that SGLT2 inhibition mitigated diabetes-induced metabolic perturbations via suppression of mTORC1 signaling in kidney tubules.¹⁶⁵ Studies in nondiabetic mice provided evidence that SGLT2 inhibition causes distinct effects on kidney metabolism, reflecting responses to partial NHE3 inhibition, and as well as urinary loss of glucose and NaCl; this included upregulation in renal gluconeogenesis and using tubular secretion of the tricarboxylic acid (TCA) cycle intermediate, α -ketoglutarate, to communicate to the distal nephron the need for compensatory NaCl reabsorption.¹³⁸

SGLT2 INHIBITION HAS DISTINCT EFFECTS ON RENAL CORTICAL AND MEDULLARY O₂ REQUIREMENTS

Mathematic modeling predicted that inhibition of SGLT2 in the diabetic kidney reduces oxygen consumption in the PCT and renal cortex, in part by lowering GFR^{141,142} (see Fig. 8.5). The predicted increase in cortical O₂ pressure and availability has been observed in a diabetic rat model using phlorizin, a dual SGLT1/SGLT2 inhibitor.¹⁶⁶ Preserving renal cortical oxygenation may be important to preserve kidney function in patients with CKD.¹⁶⁷

SGLT2 inhibition causes more equal distribution of renal transport work as it shifts glucose uptake downstream to the S3 segments (see earlier) and enhances transcellular Na⁺ reabsorption in distal segments, including the S3 segment and medullary TAL. This may further reduce the already physiologically low O₂ availability in the renal outer medulla. The latter has been proposed for SGLT2 inhibition using mathematic modeling^{141,142} and was shown *in vivo* in rats in response to acute dual SGLT2/SGLT1 inhibition by phlorizin in nondiabetic and diabetic rats.¹⁶⁶ The effect on medullary transport and oxygenation would be attenuated by the reduction in blood glucose and GFR in response to SGLT2 inhibition.^{141,142} Moreover, the proposed SGLT2

inhibitor-induced reduction in oxygen pressure in the deep cortex and outer medulla may stimulate hypoxia-inducible factors HIF-1 and HIF-2 (see Fig. 8.5). Gene knockout of SGLT2 increased the renal mRNA expression of hemoxygenase,^{5,168} a tissue-protective gene that is induced by HIF-1 α . On the other hand, activation of HIF-2 may explain an enhanced erythropoietin release from renal interstitial cells in response to SGLT2 inhibition.¹⁶⁹ Together with the diuretic effect, the latter may contribute to the observed modest increase in hematocrit and hemoglobin in response to SGLT2 inhibition. This may improve the oxygenation of the kidney outer medulla and cortex but also facilitate oxygen delivery to the heart and other organs. For a detailed discussion of the clinical use and outcomes of SGLT2 inhibitors in CKD and diabetic kidney disease, see Chapters 41 and 55. In other words and in addition to its volume effect, SGLT2 inhibition may simulate systemic hypoxia to the oxygen sensor in the deep cortex and outer medulla of the kidney, and the induced response then helps the failing heart and also the kidney.¹⁷⁰ In accordance with an overall nephroprotective effect, SGLT2 inhibitor use also reduced the risk of AKI in an analysis of >3000 patients with T2DM by ~50%.¹⁷¹ Nevertheless, caution is warranted, as excessive volume depletion and the transport shift to the outer medulla may increase the AKI risk in individual sensitive patients.

PRESERVATION OF BLOOD PRESSURE-LOWERING AND HEART FAILURE-PROTECTIVE EFFECTS OF SGLT2 INHIBITORS IN CKD DESPITE ATTENUATED ANTIHYPERGLYCEMIC EFFECTS

The amount of filtered glucose determines the glucosuric and blood glucose-lowering effect of SGLT2 inhibition. As a consequence, the antihyperglycemic effects of SGLT2 inhibitors are attenuated in patients with reduced GFR. In contrast, the blood pressure-lowering and heart failure-protective effects are preserved in patients with CKD and reduced GFR (eGFR ≥ 30 mL/min 1.73 m²)^{171a,171b} Modeling studies of CKD and nephron loss predicted that the increase in single-nephron GFR in remaining nephrons and the reduction of glucose reabsorption by SGLT2 inhibition increase paracellular Na⁺ secretion in the proximal tubule¹⁷⁰ Thus the model predicts that the chronic natriuretic and diuretic effects of SGLT2 inhibition persist in CKD. The modeling approach also predicts that the SGLT2 inhibition-induced changes in the oxygen signal at the renal sensor are preserved in CKD¹⁷⁰

ORGANIC CATIONS AND ANIONS

ORGANIC ANION TRANSPORTERS AND ORGANIC CATION TRANSPORTERS

The renal organic anion transport system, operationally characterized as the probenecid-sensitive paraaminohippurate (PAH) transport system, has been one of the best studied in physiology.¹⁷²⁻¹⁸⁰ Classically, it is presented as a proximal tubule transport system of small organic anions (e.g., PAH) bound with low affinity to plasma proteins (mainly albumin). Because, with an intact glomerular filtration barrier, albumin-bound molecules

are not filtered, they move into the peritubular capillaries. Molecules like PAH are efficiently extracted on a “first pass” by a high-capacity transport system with selectivity for organic anions. This explains why PAH clearance can be used as a measure of renal plasma flow. Operationally, the system can be blocked by the organic anion drug probenecid, which has seen considerable clinical use in the setting of hyperuricemia and to increase blood levels of organic anion drugs like penicillins and cidofovir.¹⁸¹⁻¹⁸³

The main gene responsible for this probenecid-inhibited PAH transport is an SLC transporter known as OAT1 (SLC22A6).^{184,185} Like most other members of the mammalian SLC family, it has 12 membrane-spanning segments). OAT1 and other members of the OAT subfamily are predominantly organic anion transporters, but they can also transport organic cations and zwitterions.^{186,187}

The current view of how a prototypical organic anion—PAH in this case—is taken up across the basolateral membrane (blood side) of the proximal tubule cell involves three different transporters: OAT1, the sodium-dicarboxylate cotransporter (NaDC3, SLC13A3), and the sodium–potassium ATPase (Fig. 8.7). This “tertiary” transport system is believed to operate through the following mechanisms: (1) exchange (antiport) of PAH in the plasma with intracellular α -ketoglutarate by the activity of OAT1; (2) cotransport (symport) of sodium and α -ketoglutarate into the cell by NaDC3; and (3) extrusion of sodium into the plasma, creating a sodium gradient, through the ATP-dependent action of the sodium–potassium ATPase (Na-K-ATPase). Thus organic anion transport is ATP dependent only in an indirect way and depends on two other gradients: (1) a sodium gradient generated by the Na-K-ATPase, among other factors, that enables cotransport of sodium with α -ketoglutarate into the cell by NaDC3; and (2) a high intracellular α -ketoglutarate level that is partly due to the aforementioned NaDC3 cotransport and partly contributed by aerobic mitochondrial metabolism, resulting in the generation of TCA cycle intermediates such as α -ketoglutarate. Blocking any of these processes (e.g., ouabain inhibition of the Na-K-ATPase, lithium inhibition of NaDC3, or probenecid inhibition of OAT1) markedly diminishes or even completely abolishes PAH transport via OAT1. The apical efflux pathway of organic anions taken up by OAT1 is not fully worked out, but it seems to depend in large part on MRP2 (ABCC2) and MRP4 (ABCC4), two multispecific transporters of the ABC family.¹⁸⁸

The SLC22 family has now grown to roughly 30 transporters in humans and mice^{189,190} that are involved in the transport of anionic, cationic, and zwitterionic drugs; toxins; metabolites; signaling molecules; antioxidants; dietary components; vitamins; gut microbiome products; and uremic toxins (Table 8.1). Different family members are implicated in a variety of diseases, including hyperuricemia and CKD. A new evolutionary-based classification of SLC22 has been proposed, which divides the family into a major OAT clade and a major OCT clade, which further divide into six subclades (Fig. 8.8). The assumption is that the various SLC22 subclades defined by phylogenetic relationships will lead to better functional classification and help deorphanize SLC22 transporters of unclear function. Accordingly, subclades within the OAT clade include the OAT subclade, the OAT-like subclade, and the OAT-related subclade; within

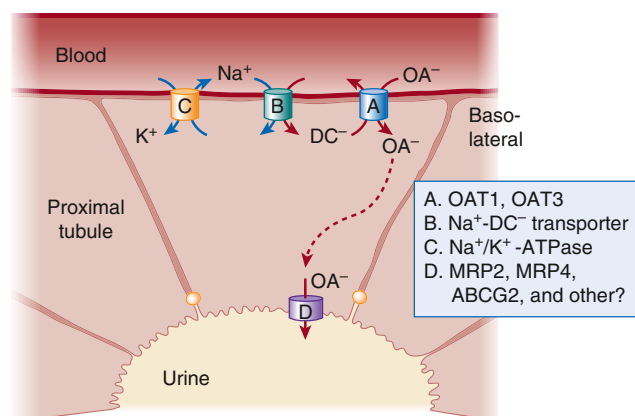


Fig. 8.7 Schematic of OAT1-mediated organic anion influx up-take as a “tertiary” transport process. Drawing of a proximal tubule cell showing OAT-mediated influx of organic anions (OA^-) from the plasma to the lumen. OAT1 (A), OA^- influx by OAT1 at the basolateral membrane via antiport of dicarboxylates (DC^-) down a gradient. OAT-mediated influx is connected to the transmembrane gradient of dicarboxylates created as a result Na^+ /dicarboxylate cotransporter and mitochondrial TCA cycle (B). This “tertiary” process depends on the ATP hydrolyzed by the Na^+ - K^+ -ATPase to create an extracellular-to-intracellular sodium gradient (C). Transport of OA^- into urinary space (D) probably occurs through a number of apical membrane transporters including the MRPs. (Modified from Nigam SK, Bush KT, Martovetsky G, et al. The organic anion transporter [OAT] family: a systems biology perspective. *Physiol Rev.* 2015;95:83–123; and Eraly SA, Bush KT, Sampogna RV, et al. The molecular pharmacology of organic anion transporters: from DNA to FDA? *Mol Pharmacol.* 2004;65:479–487.)

the OCT clade is the OCT subclade, OCT-like (OCTN) subclade, and the OCT-related subclade.^{189,191,192} Despite high expression in the kidney proximal tubule, choroid plexus, and the liver, it is worth noting that some SLC22 transporters have highly selective localization to the olfactory epithelium and brain substructures (OAT6, SLC22A20).¹⁹³ SLC22 is thus an interesting SLC family involved in the transport of anionic, cationic, and zwitterionic drugs; toxins; metabolites; signaling molecules; antioxidants; dietary components; vitamins; gut microbiome products; and uremic toxins.¹⁹⁴

Here we focus on OAT1 (SLC22A6) and OAT3 (SLC22A8), which are the major renal metabolite, drug, and toxin (including uremic toxin) organic anion transporters (see Table 8.1); OCT2 (SLC22A2), the major transporter of cationic drugs and metabolites, and URAT1 (SLC22A12), the latter being the most extensively studied of several urate transporters in the OAT subclade. Along with a number of other SLC and ABC transporters, these SLC22 transporters are the most clinically relevant and quantitatively important organic anion and organic cation transporters in the proximal tubule of the kidney.^{195,196}

OAT1 (SLC22A6) AND OAT3 (SLC22A8)

Both OAT1 and OAT3 were among the original seven drug transporters that regulatory agencies identified as important for analysis of transport of new drug entities.¹⁹⁷ This regulatory attention has perpetuated the notion that these transporters primarily transport drugs. Although it is true that the OATs transport many well-known drugs (e.g., antibiotics, antivirals, NSAIDs, diuretics) and toxins (e.g., organic mercurials and aristolochic acid), it is now clear

Table 8.1 Some SLC22 Transporter Substrates

Substrate	SLC22 Transporter			
	SLC22A6 OAT1	SLC22A8 OAT3	SLC22A1 OCT1	SLC22A2 OCT2
Nonsteroidal antiinflammatory drugs	✓	✓		
Ibuprofen				
Naproxen				
Antivirals	✓	✓		
Tenofovir				
Adefovir				
Cidofovir				
β-lactam antibiotics	✓	✓		
Ampicillin				
Benzylpenicillin				
Diuretics	✓	✓		
Bumetanide				
Furosemide				
TCA cycle intermediates	✓	✓		
Short-chain fatty acids	✓	✓		
Bile acids		✓		
Flavonoids	✓	✓		
Gut microbiome products	✓	✓	✓	✓
Organic mercurials	✓	✓		
Cisplatin			✓	✓
Metformin			✓	✓
Cimetidine		✓	✓	✓
Thiamine			✓	✓

TCA, Tricarboxylic acid.

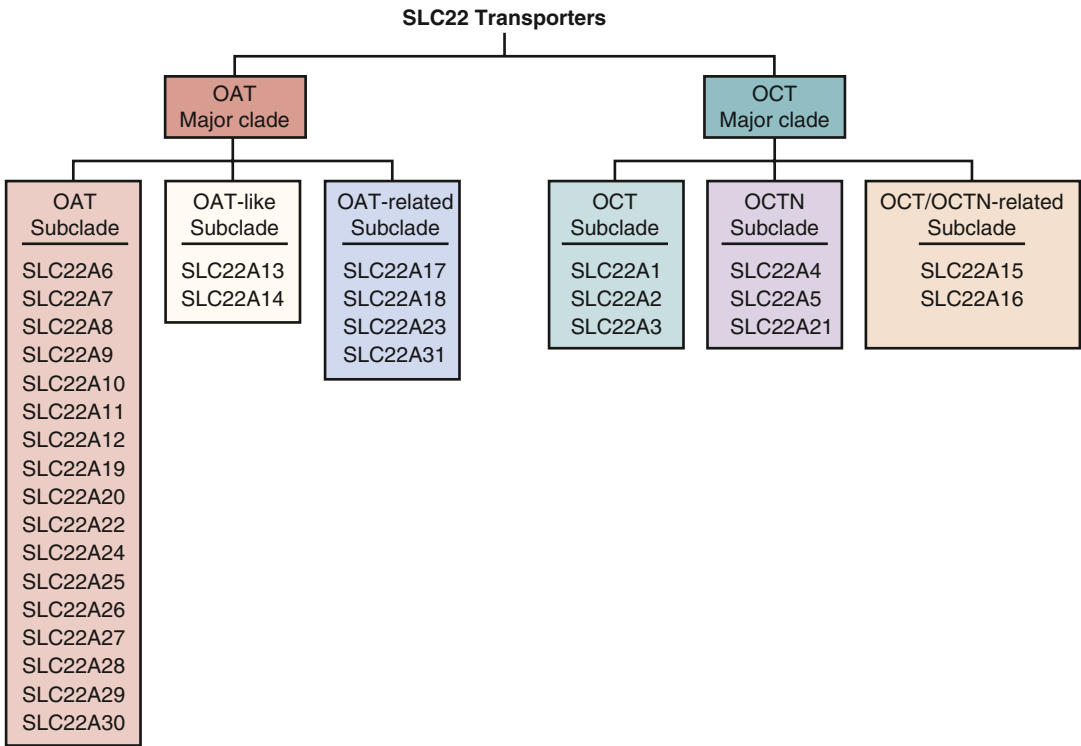


Fig. 8.8 The six subfamilies of SLC22 transporters. Evolutionary analysis indicates that SLC22 transporters are highly conserved and found in flies, worms, sea urchins, and other organisms. The SLC22 family is composed of two major clades, which are the organic anion transporter (OAT) major clade and organic cation transporter (OCT) major clade. Each of these clades is divided into three subclades, designated as OAT, OAT-like, OAT-related, OCT, OCTN (organic cation/carnitine transporter), and OCT/OCTN-related. (Modified from Nigam SK. The SLC22 transporter family: a paradigm for the impact of drug transporters on metabolic pathways, signaling, and disease. *Ann Rev Pharmacol Toxicol*. 2018;58:32.31–32.25; and Zhu C, Nigam KB, Date RC, et al. Evolutionary analysis and classification of OATS, OCTS, OCTNS, and other SLC22 transporters: structure-function implications and analysis of sequence motifs. *PLoS ONE*. 2015;10:e0140569.)

that they, as well as the other five SLC and ABC transporters highlighted by regulatory agencies, transport many endogenous metabolites, signaling molecules, vitamins, gut microbiome, and dietary products.¹⁹⁰

Indeed, the primary function of these multispecific transporters may not be the handling of drugs and toxins but rather the modulation of local and systemic metabolism and signaling.^{194,198} Much of this change in our understanding of “what drug transporters really do” is the result of “omics” analyses of knockout mice, human genome-wide association studies (GWAS), and the identification of heritable mutations that cause or modulate well-known metabolic diseases.¹⁹⁰ According to this new systems biology view (explained in more detail later), the multispecificity of these transporters for endogenous substrates enables a range of drugs and toxins to coopt these transporters expressed in the gut, liver, kidney, and many other tissues. But the pharmaceutical and commercial relevance of these transporters can create the misconception that these widely expressed and evolutionarily conserved genes primarily exist to handle synthetic drugs. Although this is apparently their key role from the perspective of clinical pharmacology and pharmacokinetics, it is becoming clear that the pharmaceutical lens is extremely limited even from the clinical point of view, since it is now evident that OAT1, OAT3, and other “drug” transporters are central to endogenous physiology and are important for understanding the pathophysiology of uremia, hyperuricemia, carnitine deficiency, and a number of genetic conditions. Importantly, short-term blockade in humans of OAT1 and OAT3 with the drug probenecid results in impressive metabolic alterations.¹⁹⁹

OAT1 KNOCKOUT AND OAT3 KNOCKOUT MICE

Oat1 knockout mouse tissue has defective uptake of the classic organic anion transporter probe, PAH, whereas the *Oat3* knockout mouse tissue has defective uptake of estrone sulfate.^{195,200} Consistent with in vitro data, the *Oat1* and/or

Oat3 knockout mice have altered in vivo or ex vivo (e.g., embryonic kidney organ cultures) handling of diuretics (e.g., loop and thiazides), antibiotics (e.g., penicillin and ciprofloxacin), a wide range of antiviral agents, and methotrexate.²⁰¹⁻²⁰⁷

Knockout mice responses to loop and thiazide diuretics provide a useful illustration (Fig. 8.9). These albumin-bound drugs in the peritubular capillaries must be transported by basolateral uptake transporters OAT1 and OAT3 in the proximal tubule, transit the cell, and exit through apical transporters (including members of the ABCC or MRP families)—all before flowing down the luminal (urinary) space to be excreted or, for the loop and thiazide diuretics, to inhibit salt reabsorption in later nephron segments. Three to five times more diuretic is required to achieve the same degree of natriuresis after deletion of either *Oat1* or *Oat3* in mice.^{195,201}

OATs are implicated in renal organic mercurial toxicity because mercury binds to glutathione and other thiol-containing compounds, many of which are “effectively” seen as organic anions by the transporter.²⁰⁸ When the *Oat1* knockout mouse was treated with high-dose mercury, the kidneys were surprisingly well protected from injury (histologically and by renal indices), consistent with the inability of the organic mercurial to be taken up by the proximal tubule due to the absence of *Oat1*.²⁰⁹

Perhaps most revealing, and somewhat unexpected, have been the results from performing metabolomics analyses on the plasma of the *Oat* knockout animals.^{195,196,210-212} These have revealed a somewhat surprising range of endogenous OAT substrates. Generally speaking, both OAT1 and OAT3 appear to play a key role in regulating the flow of organic anions through the so-called gut–liver–kidney axis.¹⁹⁶ This includes the renal handling of many compounds derived from the gut microbiome—either products of the gut microbiome or due to the action of the gut microbiome on dietary components such as phytochemicals. For example,

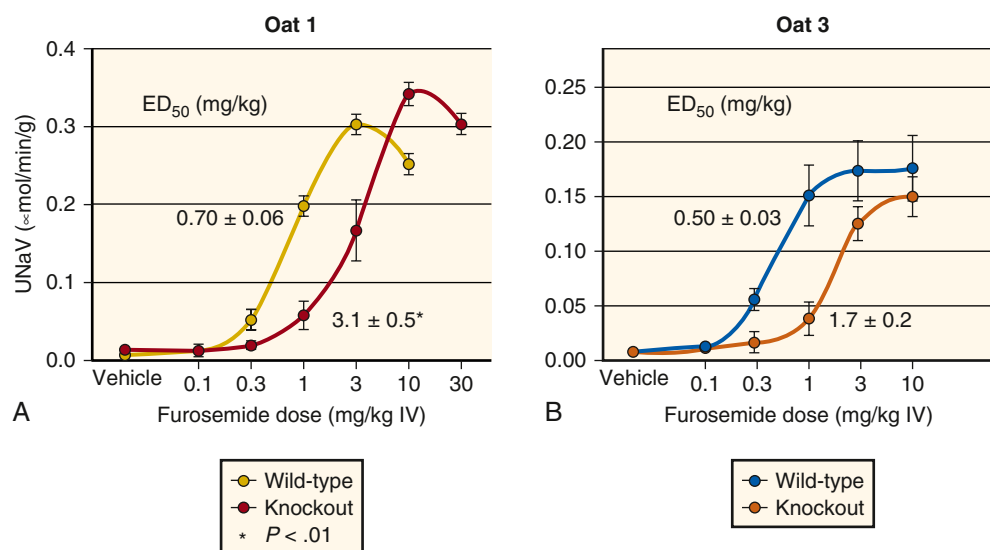


Fig. 8.9 (A, B) Marked attenuation of thiazide and loop diuretic effects in *Oat1* and *Oat3* knockout mice. In *Oat1* or *Oat3* knockout, which transport diuretics (Table 8.1), luminal sodium elimination is markedly attenuated. ED_{50} , Half-maximal effective dose; IV, intravenous; $UNaV$, urinary sodium excretion. (Modified from Eraly SA, Vallon V, Vaughn DA, et al. Decreased renal organic anion secretion and plasma accumulation of endogenous organic anions in OAT1 knockout mice. *J Biol Chem*. 2006;281:5072–5083; and Vallon V, Rieg T, Ahn SY, et al. Overlapping in vitro and in vivo specificities of the organic anion transporters OAT1 and OAT3 for loop and thiazide diuretics. *Am J Physiol Renal Physiol*. 2008;294:F867–F873.)

in the *Oat3* knockout, among the greatest changes are in flavonoids that have been acted upon by phase 2 liver enzymes (e.g., glucuronidation). This highlights the important connection between OATs (especially OAT3) with liver metabolism (via so-called drug-metabolizing enzymes) of endogenous compounds, as well as drugs and toxins. Other groups of metabolites elevated in the *Oat3* knockout include primary and secondary bile acids.¹⁹⁶ In *Oat1* and *Oat3* knockouts, fatty acids, TCA cycle intermediates, and vitamins are also elevated. A double knockout of *Oat1* and *Oat3* in rats has been reported.²¹³ Many of the analyses performed, including testing the diuretic response and the measurement of uremic toxins, were found to be consistent with results described earlier in the single knockout mice. Interestingly, the double knockout rats had uremic toxin elevation and renal dysfunction at 4 weeks that apparently resolved by 7 weeks. This compensation appears due to increased expression of another proximal tubule organic anion transporter of uremic toxins, *slco4c1*.²¹³

With respect to drugs, OAT1 and OAT3 strongly favor anionic drug substrates, but both transporters (especially OAT3) can bind a limited number of cationic/zwitterionic drugs,¹⁸⁷ such as cimetidine.¹⁸⁶ Importantly, physiochemical properties of metabolites interacting with the OATs appear different from the drug data. OAT3 metabolite substrates are larger, less polar, more chemically complex, and contain more ring structures than OAT1, though there is considerable overlap in substrates.^{196,214}

In addition, metabolic reconstructions based on changes in gene expression in the knockouts have been performed.^{212,215,216} These indicate that OATs regulate many biochemical pathways, including purine metabolism, TCA cycle, and metabolism of fatty acids, eicosanoids, amino acids, and vitamins (e.g., see top pathways affected by *Oat1* loss in knockout mice in Table 8.2). Together, the reconstructions and the metabolomics data support the view that OATs are not simply “drug” transporters but have major impact on many aspects of systemic and proximal tubule physiology.

Together, the metabolic reconstructions based on knockout omics data and the analysis of endogenous substrates (e.g., metabolites and signaling molecules) from the perspective of chemical properties call into question the oft-discussed “redundancy” of OAT1 and OAT3 in the proximal tubule. From a practical pharmacokinetic perspective, this view may still be a useful first approximation for many drugs that can interact with both OAT1 and OAT3. But considering the endogenous metabolite preferences alone (without considering drugs), a more appropriate view might be that the two transporters have distinct roles in many metabolic processes, although they work together to handle certain substrates like uric acid. On the basis of current data, renal OAT1 appears somewhat more linked to local and systemic aerobic metabolism, whereas renal OAT3 appears more linked to flow of metabolites that originate in the gut or liver (e.g., primary and secondary bile acids). Focused analyses have shown the critical role OAT1 plays in regulation of metabolism involving tryptophan, lipids, and the gut microbiome.²¹⁷⁻²¹⁹ There is also some evidence to suggest that OAT3 could modify phenotypes, for instance, in diabetic disease,²²⁰ blood pressure,²²¹ and in the setting of treatment with SGLT2 inhibitors.¹¹³

Table 8.2 Top Pathways Affected by OAT1 Loss in Knockout Mice

TCA cycle
Tyrosine metabolism
Alanine, aspartate, and glutamate metabolism
Butanoate metabolism
Arginine and proline metabolism
Tryptophan metabolism
Nicotinate and nicotinamide metabolism
Valine, leucine, and isoleucine degradation
Nitrogen metabolism
Glyoxylate and dicarboxylate metabolism
Propanoate metabolism
Glycine, serine, and threonine metabolism
Purine metabolism
Pyrimidine metabolism

TCA, Tricarboxylic acid.

Adapted from Liu HC, Jamshidi N, Chen Y, et al. An organic anion transporter 1 (OAT1)-centered metabolic network. *J Biol Chem.* 2016;291:19474–19486.

OCT2 (SLC22A2)

Unlike the OAT subclade (of the SLC22 OAT major clade), which is large, the OCT subclade (one of three subclades of the SLC22 OCT major clade) consists of three highly homologous (both protein sequence and function) transporters: OCT1, OCT2, and OCT3.¹⁸⁹ OCTs are generally held to be electrogenic uniporters.²²² Organic cation transporter 2 (OCT2) is the main renal uptake transporter on the basolateral membrane (blood side) of the proximal tubule cell that is involved in the elimination of organic cationic drugs such as metformin and *cis*-platinum.^{223,224} On the luminal (apical, urine side) of the cell, it appears that MATE (SLC47) transporter family members efflux organic cations into the urinary space.²²⁵ The primary liver OCT, OCT1—one of the drug transporters that, along with OAT1 and OAT3, has been highlighted by regulatory agencies as important for new drug testing to identify transport mechanisms—has been shown to be a thiamine transporter.²²⁶

Thus as with the OATs, OCTs may function primarily in regulating metabolite flow into and out of tissues, and as with OATs, some of the best *in vivo* functional information regarding endogenous function comes from analysis of metabolites altered in the knockout mice. On the whole, the known drug and metabolite substrates for the OCTs appear less diverse than for the OATs, and it appears that based on physiochemical property analysis of drug substrates, OCT1 and OCT2 have largely overlapping specificities, at least for drugs.¹⁸⁷

SINGLE-NUCLEOTIDE POLYMORPHISMS

The OCTs appear more polymorphic than the OATs; SNPs in OCT2 have received considerable clinical attention because, consistent with *in vitro* studies, they can affect levels of the antidiabetic agent metformin and the chemotherapeutic agent *cis*-platinum.²²⁷⁻²³⁰

Nonsynonymous coding region single-nucleotide polymorphisms (SNPs) in the OATs are uncommon compared with noncoding region SNPs.²³¹ OAT SNPs have

also been reported that affect diuretic responsiveness, mercury toxicity, antibiotic levels, and hyperuricemia.²³²⁻²³⁶ A noncoding SNP in the *OAT1* gene appears to be associated with progression of renal disease.²³⁷

Clinical Relevance: The remote Sensing and Signaling Hypothesis: A Framework for Understanding Hyperuricemia and Uremia

The OATs and OCTs have garnered a great deal of attention because of regulatory concerns due to transporter-mediated drug–drug interactions. Much more is likely on the way regarding drug–metabolite interactions (DMIs) with the advent of better techniques to analyze metabolites in tissues and body fluids. With new approaches to uremia and hyperuricemia (especially in CKD) being considered, one expects more clinical studies and trials aimed at decreasing the burden of uremic toxins and uric acid—and prolonging time to severe CKD and dialysis. In this regard, the Remote Sensing and Signaling Hypothesis should be useful for considering approaches to ameliorate perturbed interorgan and interorganismal communication via metabolites and signaling molecules, including those derived from the gut microbiome.

APICAL MEMBRANE PROXIMAL TUBULE TRANSPORTERS INVOLVED IN THE HANDLING OF ORGANIC ANIONS, ORGANIC CATIONS, AND ORGANIC ZWITTERIONS

In the proximal tubule, the OAT1, OAT3, and OCT2 function in basolateral side uptake (influx) of organic anions and cations from the blood. Although what happens to these charged organic molecules inside the cell remains poorly defined, their exit, usually in unchanged form, has become better understood in recent years. Many of the organic anions taken up by OAT1 and OAT3, for example, are effluxed across the apical membrane by members of the ABCC family, MRP2 (ABCC2) and MRP4 (ABCC4).¹⁸¹ These are not the only apical transporters of organic anions in the proximal tubule; for example, OAT4 and the well-known ABC transporters, P-glycoprotein and ABCG2, appear to play a role for certain substrates.¹⁸¹ Although in simplified representations, there is a tendency to match basolateral OAT1 and OAT3 with apical MRP2 and MRP4, it is likely that, depending on the anionic substrate taken up by OAT1 and OAT3, one or several apical transporters are involved in apical efflux into the proximal tubule lumen.

With respect to the apical efflux of organic cations, in recent years, MATEs, particularly MATE-2K, have received attention.^{181,238} MATEs are members of the SLC47 family, and it is well established that they transport many cationic drugs taken up by OCT2. Other apical transporters might also transport some OCT2 substrates, but their contribution is not well defined. It is also important to mention the organic cation/zwitterion transporters OCTN1 (SLC22A4) and OCTN2 (SLC22A5). Although they are sometimes listed along with drug transporters, their endogenous substrates—ergothioneine and carnitine, respectively—are well established.^{239,240} Indeed, mutations in OCTN2

Table 8.3 Uremic Toxins Accumulating in OAT1 and/or OAT3 Knockout Mice

Indoxyl sulfate	Indolelactate
p-Cresol sulfate	Kynurenate
Hippurate	Putrescine
CMPF	Uric acid
Phenyl sulfate	Creatinine
Xanthurenate	

CMPF, 3-Carboxy-4-methyl-5-propyl-2-furanpropanoate.

Adapted from Wikoff WR, Nagle MA, Kouznetsova VL, et al.

Untargeted metabolomics identifies enterobiome metabolites and putative uremic toxins as substrates of organic anion transporter 1 (OAT1). *J Proteome Res.* 2011;10:2842–2851; and Wu W, Bush KT, Nigam SK. Key role for the organic anion transporters, OAT1 and OAT3, in the in vivo handling of uremic toxins and solutes. *Sci Rep.* 2017;7:4939.

cause systemic carnitine deficiency, which can lead to severe cardiac and skeletal myopathy. Ergothioneine, on the other hand, is considered an important antioxidant. Nevertheless, OCTNs have some ability to interact with cation drugs.

RENAL TRANSPORT OF SPECIFIC ORGANIC SUBSTRATES IN DISEASE

TRANSPORT OF UREMIC TOXINS AND GUT MICROBIOME-DERIVED SMALL MOLECULES

The importance of OATs in renal handling of gut microbiome-derived metabolites merits further discussion. Many of the metabolites accumulating in the *Oat* knockouts, such as indoxyl sulfate, kynurenine, p-cresol sulfate, and hippurate, are among the sets of gut microbiome-derived small molecules (organic anions); they are also frequently implicated as “uremic toxins”^{210,211} (Table 8.3).

The list of small molecules implicated in uremic toxicity is long and much debated.^{241,242} These uremic toxins are thought to play a role in many tissue and organ toxicities and dysfunctions that occur in the uremic syndrome associated with severe CKD. It is unlikely that any single uremic toxin on the list is the key to all the manifestations of the uremic syndrome, although there is growing evidence that certain uremic toxins play a role in particular tissue toxicities. These include TMAO (trimethylamine-N-oxide), which has been implicated in cardiovascular toxicity,^{238,243} and indoxyl sulfate, which has been implicated in multiple aspects of the uremic syndrome.²⁴⁴

Some of these uremic toxins (e.g., indoxyl sulfate) may play a role in the actual progression of renal disease, presumably in part via OAT-mediated uptake into proximal tubule cells.^{211,245} The pathophysiology of uremia is beyond the scope of this chapter, and readers are referred instead to Chapter 51. But it is important to emphasize the growing appreciation of the role of OAT1 and OAT3 in regulating levels of many molecules considered uremic toxins. Thus it will be interesting to determine whether factors that affect the expression or function of OAT1 and OAT3 alter the time at which the uremic syndrome develops in CKD or the progression of renal disease.

OCT2 also appears to be a transporter of certain cationic uremic toxins, notably TMAO, a molecule accumulating in CKD that is associated with cardiovascular disease.²⁴⁶ The apical transporters of organic anions, the MATEs, appear to be involved in the efflux of TMAO. Of note, the deletion in mice of OAT3, which can transport a few organic cations despite being an organic anion transporter, results in elevated levels of TMAO, although it is not clear whether TMAO is actually transported by OAT3.²¹¹

URIC ACID TRANSPORT

Uric acid is a product of purine metabolism and a weak organic acid with a pKa of 5.5–5.75, existing primarily as urate at physiological pH.²⁴⁷ The formation of uric acid is catalyzed by the enzyme xanthine oxidase/xanthine dehydrogenase from xanthine and hypoxanthine.²⁴⁷ In many organisms, uric acid is oxidized by the enzyme uric acid oxidase (uricase) into allantoin;²⁴⁷ however, mammalian primates demonstrate a gradient of reduced uricase function culminating in complete loss of function of the uricase gene (UOX) in the great apes.^{248,249} Humans have accumulated three separate mutations in UOX supporting the argument of strong selective pressure for the loss of uricase function.^{248,249} The hypothesized adaptive advantage gained with the loss of uricase function remains controversial, though roles of urate as an antioxidant and regulator of beneficial metabolic pathways have significant support.^{250,251} The disadvantages for human health, however, are clear: Increased serum urate levels result in elevated risk of monosodium urate crystal precipitation and gout.^{247,252} In addition to gout, hyperuricemia (clinically defined elevated serum urate levels) and urate are independently associated with the primary drivers of complex human diseases including renal diseases, hypertension, cardiovascular disease, and metabolic syndrome.^{247,253,254} The root of all urate-related pathology is a derangement of the careful balance between urate production, primarily in the liver, and excretion, maintained by epithelial transport systems of the liver, intestines, and kidney.

The kidney is responsible for 70% of total urate excretion^{255,256}; however, the homeostatic balance between excretion of a nitrogenous waste product and prevention of urate crystal deposition has resulted in a complicated distribution of urate absorption and secretion transporters in the proximal tubule.^{247,257} Urate is freely filtered by the glomerulus,²⁵⁸ yet the final fractional excretion of urate is between 5 and 15%,^{247,253,259–261} suggesting active urate transport along the tubule. Early transport studies using radiolabeled uric acid²⁶² and renal epithelial brush border membrane vesicle studies²⁵⁶ suggested components of active transport and active secretion along the three segments of the proximal tubule.²⁴⁷ Our current understanding of urate handling suggests that up to 95% of the filtered urate is rapidly reabsorbed in the S1 segment of the proximal tubule as part of the initial bulk reabsorption of many organic anions.^{247,259,261} Next, there is active secretion of urate back into the tubule lumen, as much as 50% of the original filtered load,^{263–265} occurring in the S2–S3 segments of the proximal tubule. Proximal urate handling is completed with a second phase of reabsorption in the latter S2 and S3 segments.^{259,261} Though much of this scheme was originally

proposed based on human patient-based observational trials, recent identification of the urate transporter genes, mouse knockout studies, and mapping of the transporter gene expression and protein localization has confirmed this complicated urate handling physiology of the proximal tubule in humans.²⁴⁷

Molecular basis of urate transport

SLC22A12 (URAT1) was the first identified urate transporter.²⁶⁶ A large number of GWAS of either serum urate or gout risk have since identified more than 183 genetic loci^{254,267,268} including a large number of loci containing transporter genes. Most code for transport proteins expressed in the proximal tubule of the kidney including (gene name/protein name; proximal tubule region highest gene expression): *ABCG2* (ABCG2/BCRP, S2–3), *SLC2A9* (SLC2A9/Glut9; S1–3), *SLC22A12* (URAT1; S2–3), *SLC22A6* (OAT1; S1–2), *SLC22A7* (OAT2; S3), *SLC22A11* (OAT4; S1 and S3), *SLC17A1* (NPT1; S1), and *SLC17A3* (NPT4; S1),²⁴⁷ with additional evidence for *SLC22A9*, *SLC22A8* (OAT3), *SLC22A13* (OAT10), *ABCC4* (MRP4) in select populations.^{269,270} Of all the urate-associated genes, variants in just three, *ABCG2*, *SLC2A9*, and *SLC22A12*, contribute the largest portion of the measured variability in serum urate levels^{254,271} marking alterations in urate transport as the major driver of genetic urate-related pathologies. Follow-up functional analysis has demonstrated that both *ABCG2*^{272,273} and *SLC2A9*,^{269,274,275} in addition to URAT1, are transporters with significant urate affinity with *ABCG2* being the elusive secretory uric acid secretory transporter.

The protein product of *SLC22A12*, URAT1, has a K_m for urate of $371 \pm 28 \mu\text{M}$,²⁶⁶ demonstrating higher affinity for urate than URAT1 of other species, a phylogenetic divergence that appears to be correlated with the loss of uricase function in the ape lineage.²⁷⁶ This suggests that URAT1 may function to limit urate levels in the nephron. In vitro studies using *Xenopus* oocytes revealed that Cl[−] and organic anions including lactate, nicotinate, succinate, pyrazinecarboxylic acid (PZA), and orotate-*cis* (extracellular) inhibit urate absorptive transport.^{266,277} Many of them, like PZA and lactate, are also substrates for URAT1 and *trans*-stimulate urate transport.^{266,277} The functional coupling of urate reabsorption to intracellular metabolites and Krebs cycle intermediates suggests that URAT1 function is an important component of metabolic pathways in the proximal tubule.²⁷⁸ Many uricosuric drugs target URAT1, inhibiting urate reabsorption (e.g., probenecid and lesinurad). SGLT2 inhibitors have a significant uricosuric effect as well.^{153,279} The mechanism of the uricosuric effect of SGLT2 inhibitors is not understood, but work from mouse models and clinical studies support a functional coupling between glucose reabsorption and URAT1,^{153,278} indicating glucose or sodium may be influencing urate levels.

SLC2A9 variants make the largest contribution to serum urate variability, but the functional relationship is complicated, with variants associating with both higher or lower urate levels.^{271,280} Part of the complication is due to the expression of two different protein products or isoforms of *SLC2A9*, which may have divergent tissue expression and localization.^{260,281} In addition, *SLC2A9* is expressed in all of the key urate-handling tissues: liver, intestine, and kidney

and may be part of the secretory path in some (intestine) and reabsorption in others (kidney).^{260,281,282} Unlike URAT1, SLC2A9-mediated urate transport is electrogenic (both isoforms) with cell depolarization enhancing urate flux.^{283,284} The K_m for urate is approximately 890 μM , with a high capacity for urate transport.²⁶⁸ SLC2A9 is a member of the GLUT gene family (GLUT9), but transport studies in *Xenopus* oocytes have not found any evidence that SLC2A9 facilitates glucose or fructose transport. However, intracellular glucose was shown to increase the rate of SLC2A9-mediated urate efflux.^{284,285} In addition to the human genetic studies, mouse models have also implicated the importance of SLC2A9 in renal urate excretion, with nephron-specific *Slc2a9* knockout mice displaying a significant increase in urate excretion.^{279,281} Interestingly, the localization of the SLC2A9 in mice is highest in the DCT/CNT, not proximal tubule, suggesting a possible role for distal urate transport in regulating overall urate excretion and homeostasis.^{279,281,284}

In contrast to URAT1 or SLC2A9, ABCG2 is a secretory urate transporter expressed in many epithelial tissues including the liver, intestine, and kidney.^{247,252,260} ABCG2 is a member of the multidrug transporter gene family and has been studied extensively for its role in drug transport^{273,286}; however, its primary physiologic role appears to be as a high-capacity urate transporter of the kidney and intestine.²⁷³ Variants in ABCG2 are the largest genetic contributor to gout risks,^{271,273} with the majority of the genetic risk from one common missense variant (Q141K, allele frequency 0.11–0.31²⁸⁷). This missense variant is a partial loss-of-function mutation resulting in instability of the nucleotide-binding domain (NBD) and increased endoplasmic reticulum-associated protein degradation (ERAD).²⁸⁸ Data from a human interventional trial of individuals with the Q141K ABCG2 variant and from a knock-in mouse model of the same *Abcg2* variant confirmed ABCG2's key role in both renal and intestinal secretion^{247,260} but also surprisingly revealed almost complete loss of function in the gut yet mild pathology in the kidney.^{247,260} The same study quantified the role of renal urate excretion in mice and humans in regulating serum urate and found that it was indistinguishable, supporting the mouse as a model for studying human renal urate biology.^{247,260}

Sex differences in urate transport

Genetic and epidemiologic studies have quantified a dramatic sex difference in gout and hyperuricemia risk, with males being four times more likely affected than women.²⁵³ As women age, their risk for urate-related disease increases, with a fivefold increase in gout risk after menopause,²⁵³ supporting a powerful role for sex hormones in regulating serum urate levels. Studies have shown differences in renal urate transporter expression that translate to fundamental sex differences in renal urate handling (females displaying a significantly higher maximum FEUA with a urate load).^{253,260} GWAS has shown specific sex differences in the contribution to disease risk for variants in *ABCG2* and *SLC2A9*,^{254,288} differences recapitulated in mouse models for the same human variants,²⁶⁰ and has led to identification of a key urate-related transcription factor, *HNF4A*,²⁵⁴ a

possible driver in the sex differences in renal transport architecture.²⁹⁰

CREATININE TRANSPORT

A number of members of the SLC22 family (containing OATs and OCTs) transport creatinine. These include OCT2, OAT2, OAT3, and possibly OAT1.^{239,291,292} All of these are considered key transporters of organic cation drugs, organic anion drugs, and/or zwitterionic drugs. Their relative importance in renal creatinine handling is debated, but it is likely that all play some role in the renal handling of creatinine, with OCT2 being particularly important. The issue is clinically relevant because a number of drugs (e.g., trimethoprim) are thought to “artificially” create the impression of renal dysfunction when creatinine is the primary measure used because of drug–metabolite interactions at the level of the transporter.¹⁷⁷

THE REMOTE SENSING AND SIGNALING THEORY

It is evident that multispecific “drug” transporters of both the SLC and ABC transporter family are critical in the local and systemic regulation of levels of a huge array of metabolites.^{189,190,198,194,240} The Remote Sensing and Signaling Theory¹⁹³ argues that the physiologic role of multispecific “drug” transporters and their close (often mono- or oligo-specific) relatives is the regulation of “remote” interorgan and interorganismal communication via metabolites and signaling molecules.

The system is hypothesized to be actively regulated through transcriptional mechanisms (e.g., nuclear receptors) and posttranslational mechanisms (e.g., kinases regulating transporter internalization or PDZ domain association). It is envisioned to work in parallel with the neuroendocrine, growth factor cytokine, and autonomic systems.^{293,294} This theory seems to be particularly relevant to renal disease.^{295–297} An injured organ—the malfunctioning kidney in CKD—is involved in a kind of “organ cross-talk” with the intestine, whereby intestinal ABCG2 transporters, and possibly others, appear to “take over” from many renal SLC uric acid transporters as tubular function declines (Fig. 8.10). As discussed earlier, it seems likely that high urate, certain uremic toxins, or both—accumulating as a consequence of renal disease—are thought to induce expression of intestinal transporters.²⁹⁸

The uremic syndrome can be partly viewed as a disorder of the remote sensing and signaling system due to “pathologic” transporter-mediated remote interorgan and interorganismal communication of protein-bound “uremic toxins.” These uremic toxins include signaling molecules that bind G-protein-coupled receptors (e.g., kynurenine) and nuclear receptors (e.g., indoxyl sulfate). The Remote Sensing and Signaling Theory provides a different lens on the numerous biochemical and cellular aberrations found in the uremic syndrome, which affect many tissues and body compartments. Moreover, the source of many uremic toxins in the body is the gut microbiome—the result of interorganismal communication—and these toxins affect signaling and metabolism in multiple organs before, in many cases, being eliminated by proximal tubule multispecific

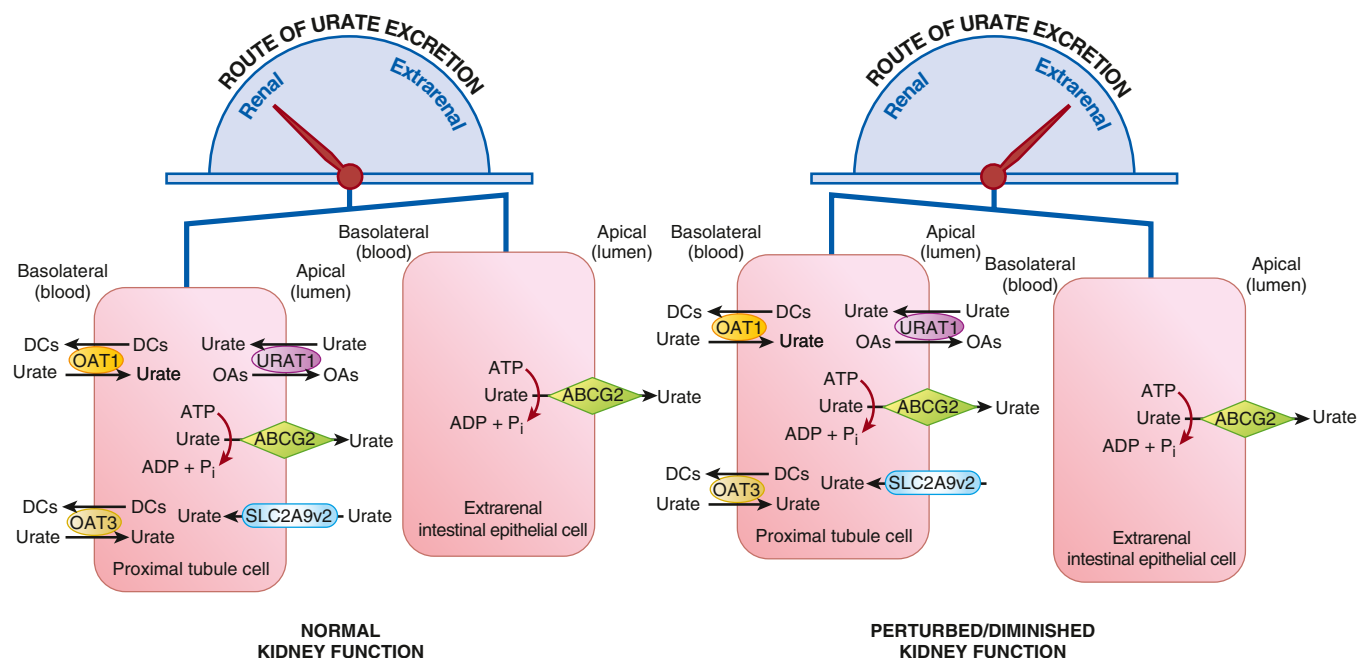


Fig. 8.10 In chronic kidney disease, the intestinal role of uric acid transporters (mainly ABCG2) becomes much more important. Schematic of urate excretion in the setting of normal renal function versus that in which there is diminished renal function. In normal situations, the majority of urate excretion (~70%) is performed mainly in the proximal tubule of the kidney and is mediated by a number of transporters found on the apical and basolateral membranes of the proximal tubule cell. The illustration depicts only some of the transporters involved and includes members of the SLC and ABC families of membrane transporters. On the basolateral surface, the uptake of uric acid from the blood is mediated mainly by OAT1 (SLC22A6) and OAT3 (SLC22A8). These solute carrier transporters exchange dicarboxylates (DCs) for urate, resulting in the net movement of this organic anion into the proximal tubular cell. At the apical surface, a number of transporters including ABCG2 (depicted here), NPT1 (SLC17A1), and NPT4 (SLC17A3), as well as ABCG family members, work to secrete urate into the tubular lumen for excretion via the urine. The relative importance of the aforementioned apical transporters in urate secretion remains unclear. Transport of urate via ABCG2 is driven through ATP hydrolysis. Several other apical membrane transporters are well established in the reabsorption of urate including URAT1 and SLC2A9v2. URAT1, or SLC22A12, exchanges intercellular organic anions for urate, resulting in the movement of uric acid back into the proximal tubule cell. Meanwhile, under normal physiologic conditions, up to 30% of uric acid is excreted via extrarenal transporters, believed to largely be driven by ABCG2 expressed in intestinal epithelial cells. The results of the analysis of human data—and physiologic data from rodent models with renal dysfunction—support the view that urate transport by ABCG2, most likely in intestine, compensates for poor renal urate handling in the setting of diminished kidney function. (Modified from Bhatnagar V, Richard EL, Wu W, et al. Analysis of ABCG2 and other urate transporters in uric acid homeostasis in chronic kidney disease: potential role of remote sensing and signaling. *Clin Kidney J.* 2016;9:444–453.)

“drug” transporters, such as OAT1, OAT3, OCT2, and the MATEs.

DRUG–METABOLITE INTERACTIONS

As more is learned about the endogenous substrates of OATs, OCTs, and other drug transporters, and as more is learned through metabolomics and metabolic reconstructions (such as those described earlier) regarding the regulation of tissue-specific metabolism by these transporters, as well as the broader function of these transporters in endogenous metabolism and organ crosstalk (as described by the Remote Sensing and Signaling Theory), it will become possible to consider new ways to modulate complex metabolic diseases like uremia and to understand the implications of drug–metabolite interactions (DMI) beyond simple competition at the level of the transporter itself.^{177,215} Much work needs to be done in the area of DMI where competition for transport by the drug may affect a wide range of metabolic pathways; this would seem particularly relevant in the setting of moderate CKD, in which the affected metabolites would likely include certain uremic toxins.¹⁷⁷

AMINO ACIDS

PHYSIOLOGY OF RENAL AMINO ACID TRANSPORT

Rare inherited defects of renal amino acid transport have been instrumental in our understanding of renal metabolite reabsorption.^{299,300} Four disorders associated with apical amino acid transporters define four major renal transport pathways, namely a transporter for neutral amino acids (mutated in Hartnup disorder), a transporter for cationic amino acids and cystine (mutated in cystinuria), a transporter for anionic amino acids (mutated in dicarboxylic aminoaciduria), and a transporter for glycine and proline (mutated in iminoglycinuria)³⁰¹ (see Chapter 44 for a detailed discussion of these inherited disorders). With the exception of iminoglycinuria, aminoacidurias also affect intestinal transport. Aminoacidurias affecting apical transport have little effect on other organs, indicating that expression of these transporters is largely specific to the apical membrane in the kidney and intestine, although some renal amino acid transporters are also found in the brain. Transport across the basolateral membrane, however, has been more challenging to delineate. More than 98%

of all filtered amino acids are reabsorbed in the proximal tubule.³⁰² Some differences in transporter expression are observed between the PCT and the PST (Fig. 8.11).

MOLECULAR BIOLOGY OF RENAL AMINO ACID TRANSPORTERS

Renal epithelial amino acid transporters are found in a variety of SLC families^{301,303} (Table 8.4). An overview of the amino acid transporter endowment of renal epithelial cells is depicted in Fig. 8.11.

TRANSPORTERS FOR NEUTRAL AMINO ACIDS

APICAL TRANSPORTERS

The presence of a dominant transporter for neutral amino acids can be inferred from the aminoaciduria observed in Hartnup disorder, which is restricted to neutral amino acids.³⁰⁴ The affected gene encodes the amino acid transporter B⁰AT1 (Broad neutral (0) Amino acid Transporter 1, SLC6A19).^{305,306} Although Hartnup disorder shows simple recessive inheritance and therefore is monogenic, the transporter protein requires association with one of two ancillary type I transmembrane proteins to traffic to the apical membrane and to be fully functional,³⁰⁷ collectrin (TMEM27) in the kidney³⁰⁸ and angiotensin-converting enzyme 2 (ACE2) in intestine.³⁰⁹ Both proteins

are type I transmembrane (TM) proteins with a single TM domain. Although ACE2 is expressed in the kidney, its expression levels in the proximal tubule are too small to make a significant contribution to B⁰AT1 surface expression. B⁰AT1 and TMEM27 are both expressed in the PCT.³⁰⁸ Consistent with the monogenic inheritance, mutations in TMEM27 have not been observed in Hartnup disorder and would be expected to show an amino acid transporter defect in the kidney but not intestine.^{310,311} Thus far more than 20 different causative mutations have been identified in the *SLC6A19* gene.³¹² Interestingly, rare variants in *SLC6A19* have been associated with low serum creatinine levels and may have a kidney-protective effect or affect creatinine synthesis.³¹³ This is consistent with data in mice where aristolochic acid-induced nephropathy is attenuated by the lack of SLC6A19.³¹⁴ B⁰AT1 transports all neutral amino acids in symport with 1Na⁺,³¹⁵ with a preference for branched-chain amino acids (BCAA) and methionine, followed by large hydrophilic and aromatic amino acids.³¹⁶ In vitro systems, proline and tryptophan are poor substrates of B⁰AT1, but in vivo data suggest a significant contribution to the transport of both amino acids.³¹⁷ Mice lacking B⁰AT1 replicate the human aminoaciduria of Hartnup disorder but do not show any additional pathology (Table 8.5). In addition to the low-affinity B⁰AT1 transporter in the PCT, functional studies suggest the presence of a high-affinity transporter for

Table 8.4 Human Tubular Amino Acid Transporters Properties and Distribution of Human Renal Tubular Transporters

Amino Acid Transporter	SLC	PCT/PST	Substrates	Affinities	Disease	Structure Class
Apical						
B ⁰ AT1	SLC6A19	PCT	All neutral	800–15,000 μM	Hartnup disorder (OMIM 234500)	LeuT
B ⁰ AT2	SLC6A15	PCT	BCAA, Met, Pro	50–200 μM	n.r.	LeuT
Collectrin	TMEM27	PCT	N/A	Ancillary	n.r.	1TM
rBAT	SLC3A1	PCT < PST	N/A	Ancillary	Cystinuria (OMIM 220100)	1TM
b ⁰ +AT	SLC7A9	PCT > PST	Arg, Lys, Orn, CysC, Met, Leu, Ala	100 μM	Cystinuria (OMIM 220100), isolated cystinuria (OMIM 238200)	LeuT
EAAT3	SLC1A1	PCT < PST	Glu, Asp, CysC	20–80 μM	Dicarboxylic amino aciduria (OMIM 222730)	GlT
AGT1	SLC7A13	PCT = PST	Glu, Asp, CysC	20–60 μM	n.r.	LeuT
PAT2	SLC36A2	PCT			Iminoglycinuria (OMIM 242600), Hyperglycinuria (OMIM 138500)	LeuT
SIT	SLC6A20	PCT < PST			Iminoglycinuria (modifier) (OMIM 242600)	LeuT
TauT	SLC6A6	PCT, PST			n.r.	LeuT
Basolateral						
LAT2	SLC7A8	PCT			n.r.	LeuT
4F2hc	SLC3A2	PCT = PST		Ancillary	Lethal	1TM
y ⁺ LAT1	SLC7A7	PCT = PST			Lysinuric protein intolerance (OMIM 222700)	LeuT
TAT1	SLC16A10	PCT			n.r.	MFS
LAT4	SLC43A2	PCT = PST	BCAA, Met, Phe	5000 μM	n.r.	MFS

BCAA, Branched-chain amino acids; CysC, cystine; 1TM, single transmembrane-helix protein; MFS, multifacilitator superfamily; n.r., not reported; OMIM, Online Mendelian Inheritance in Man; PCT, proximal convoluted tubule; PST, proximal straight tubule. Links to Diseases Refer to the OMIM Database.

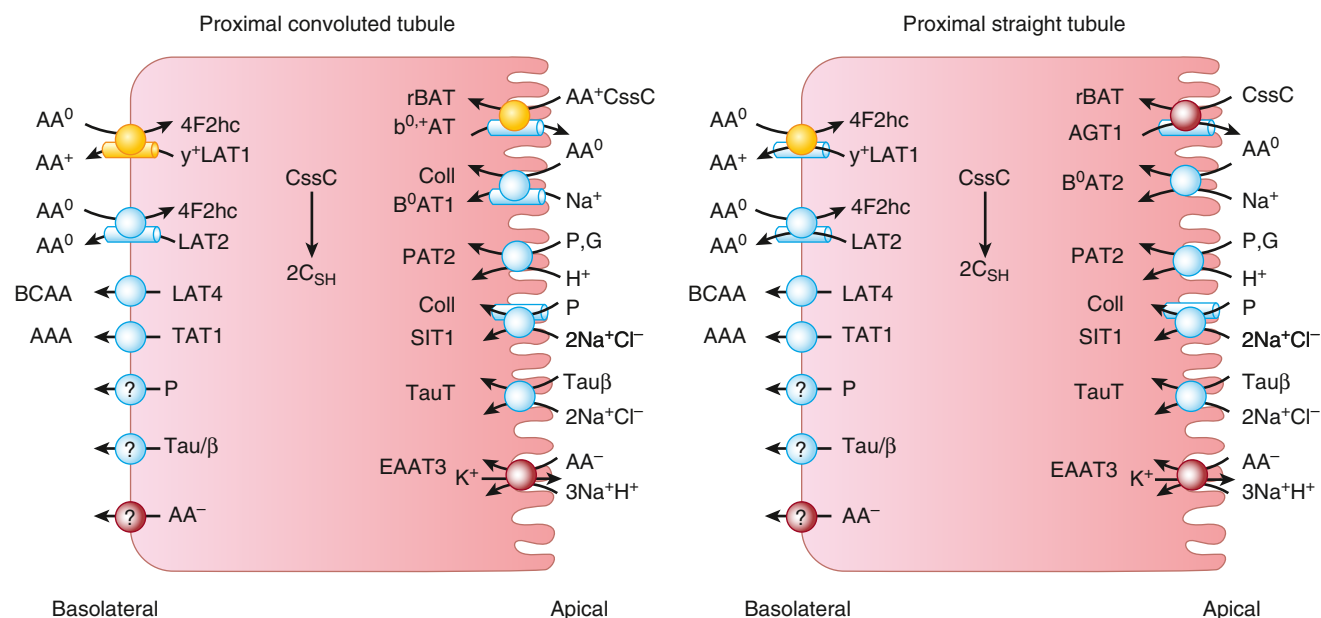


Fig. 8.11 Amino acid transporters in the proximal tubule. The common names of amino acid transporters in the proximal convoluted tubule and proximal straight tubule are shown next to each transporter. Transporter requirements for ancillary subunits are indicated by a tubelike structure. Transporters for anionic amino acids are shown in red; transporters for cationic amino acids are shown in yellow. AA^0 , neutral amino acids; AA^+ , cationic amino acids; AA^- , anionic amino acids; AAA, aromatic amino acids; BCAA, branched-chain amino acids; C_{SH} , cystine; CysC, cystine; G, glycine; P, proline; Tau/β, taurine and β-amino acids.

Table 8.5 Physiologic Characteristics of Knockout Mice of Renal Amino Acid Transporters

Transporter	Plasma AA	Urine AA	Renal Pathology	Other Features
rBAT	Normal	+++ Lys, Arg, Orn, CysC	Nephritis, kidney stones	None
$b^{0,+}AT$	+: His, Ser, Glu/n	+++ Lys, Arg, Orn, CysC, + Glu/n	Glomerular fibrosis, nephritis, kidney stones	None
B^0AT1	Normal	+++ Neutral AA	None	Propensity for colitis, improved glycemic control
B^0AT3	Normal	+++ Gly ++ Ala, Val, Leu, Ile, Met, Ser, Thr, Gln, Phe, Tyr	Reduced serum creatinine None	Stress-induced increase of blood pressure
TAT1	+: Phe, Tyr, Trp -: Gly, Ala, Met, Ser, Thr, Asn, Gln HPD +: Phe, Tyr, Trp	+: Phe, Tyr, Trp HPD +: Val, Ile, Leu, Thr, Gln, His, Phe, Tyr, Trp	None	None
LAT4	-: Ala, Pro, His, Ser	Not available	None reported	Liver inflammation, malnutrition
LAT2	+: Gly, Ala, Ser, Thr, Gln, Val, Lys	+: Gly, Ser, Thr, Gln, Leu, Val	None	None
y^+LAT1	Not available	+++ Lys, Arg, Orn	None	Failure to thrive, intrauterine growth restriction
TauT	-: Taurine -: Glu	+++ Taurine	Enlarged kidney, glomerulosclerosis, nephropathy	Muscle weakness, cardiomyopathy, retinal degeneration, hearing loss, chronic liver disease

CysC, Cystine; Glu/n, glutamate or glutamine; HPD, high-protein diet; +, elevated, ++ significantly elevated, +++ highly elevated; -, reduced, --, significantly reduced.

neutral amino acids in the PST. A candidate could be B⁰AT2 (SLC6A15), which is expressed at low levels in the proximal tubule.³¹⁸ B⁰AT2 has a narrower substrate specificity than B⁰AT1, showing a strong preference for BCAA and methionine with substrate affinities <100 μM .³¹⁹

Specific apical transporters for proline and glycine

Proline and glycine have unusual physicochemical properties, which cause these amino acids to be inefficiently transported by more broadly specific amino acid transporters. Glycine is lacking a side chain, reducing its affinity to side-chain binding pockets, whereas proline has a secondary amino group and restricted flexibility. As a result, proline is a poor substrate for B⁰AT1 and is not recognized by the basolateral neutral amino acid transporter LAT2 (see later). A common transporter for glycine, proline, and hydroxyproline in humans is supported by two lines of evidence. First, in the rare disorder iminoglycinuria, all three amino acids are found in the urine.³¹⁷ Second, prolinemia, when passing the renal threshold, causes prolinuria, hydroxyprolinuria, and glycinuria.³²⁰ Although iminoglycinuria is an autosomal recessive disorder, it shows clear signs of genetic complexity.³²¹ For instance, some cases of iminoglycinuria show malabsorption of proline in the intestine. Moreover, in some cases, heterozygotes are normal, whereas in other pedigrees, hyperglycinuria is observed. These cases can all be explained by the combined action of proton–amino acid transporter 2 (PAT2, SLC36A2), the system IMINO transporter SIT1 (SLC6A20), and the general neutral amino acid transporter B⁰AT1 (SLC6A19).³¹⁷ Homozygous mutations in PAT2 account for iminoglycinuria, whereas heterozygous mutations cause selective hyperglycinuria. SIT1 mutations contribute to the iminoglycinuria phenotype and also explain a sporadically observed reduced proline absorption in the intestine. B⁰AT1 provides the baseload for proline and glycine reabsorption, explaining why the extent of aminoaciduria is well below the filtered amounts.³¹⁷ The related glycine and alanine transporter B⁰AT3 (SLC6A18) is only functional in mouse^{322,323} (see Table 8.5) but not in higher mammalian species, where its function has been replaced by PAT2.

PAT2 (SLC36A2) has been identified as the specific transporter for small neutral amino acids, specifically alanine, glycine, and proline. Single N-methylation of amino acids is tolerated such as in sarcosine or proline. PAT2 is mutated in all cases of iminoglycinuria and is expressed in the proximal tubule of the kidney.³¹⁷

SIT1 (SLC6A20) accepts only amino acids with secondary, tertiary, or quaternary amines, such as proline, sarcosine, betaine, and methylaminoisobutyric acid.^{324,325} SIT1 is an Na⁺ and Cl[−] dependent transporter, transporting 1 substrate, 2Na⁺, and 1Cl[−] ion. Expression of the transporter at the cell surface requires collectrin (TMEM27).³²⁶

B⁰AT3 (SLC6A18) preferentially transports alanine and glycine.³²³ SLC6A18 knockout mice have hyperglycinuria and slightly elevated urine levels of other neutral amino acids,³²² but humans with homozygous mutations in SLC6A18 are normal because even the normal allele is nonfunctional.^{317,327} The mouse transporter requires collectrin coexpression for trafficking and also for its catalytic function.³²⁷

Specific apical transporters for β -amino acids

This group of amino acids consists of taurine, β -alanine, and its homolog gamma-aminobutyric acid (GABA). Plasma concentration of GABA is low (about 0.1 μM), while taurine levels are significantly higher (about 50 μM). Both amino acids are often accepted by the same transporters, but in the kidney, specific GABA transporters are observed as well. Reabsorption of these amino acids is mainly mediated by the taurine transporter TauT (SLC6A6).³²⁸ PAT2 shows weak affinity for these substrates but is probably physiologically irrelevant. TauT translocates taurine with a K_M of 20 μM in a process that involves the cotransport of 2Na⁺ and 1Cl[−]. Urinary taurine excretion in TauT-deficient mice reaches the filtered amounts.³²⁹

BASOLATERAL TRANSPORTERS

The heteromeric amino acid transporter 4F2hc-LAT2 (SLC3A2-SLC7A8) is a basolateral Na⁺-independent transporter broadly specific for neutral amino acids that functions as an antiporter.^{311,330,331} It accepts all neutral amino acids except proline. A uniporter has been identified for aromatic amino acids (TAT1, SLC16A10),³³² but this transporter cannot explain the net flux of amino acids across the renal epithelium, as *Tat1* knockout mice show elevated levels of aromatic amino acids in the urine but normal levels of other amino acids.³³³ Apart from aromatic amino acids, TAT1 also transports the N-methylated derivatives of these amino acids and L-dopa. LAT4 (SLC43A2), which belongs to a separate transporter family than 4F2hc-LAT2, allows efflux of BCAA, methionine, and phenylalanine, thus covering large neutral amino acids.³³⁴ A specific efflux pathway for small neutral amino acids has not been identified and thus is likely to occur through cooperation between LAT2, TAT1, and LAT4, which is supported by the appearance of small neutral amino acids in the urine of *Lat2* knockout mice.³³⁵

The kidney contributes to gluconeogenesis and pH regulation. The glutamine, asparagine, and histidine transporter SNAT3 (SLC38A3) is located in the basolateral membrane but is typically expressed at low levels. During chronic acidosis, its mRNA is upregulated and glutamine is imported. The resulting glutamate is further deaminated to 2-oxoglutarate, which serves as a substrate for gluconeogenesis^{336,337} (see also Fig. 8.4). The related transporter SNAT (SLC38A5) is also expressed in the kidney and may contribute to glutamine uptake.³³⁸

There is no clear pathway for the basolateral exit of proline and taurine.

TRANSPORTERS FOR CATIONIC AMINO ACIDS AND CYSTINE

APICAL TRANSPORTERS

Cationic amino acids and cystine are transported in the proximal tubule by the heteromeric transporter rBAT/b⁰+AT.³³⁹ Homozygous (or compound heterozygous) mutations in the rBAT-encoding gene *SLC3A1*, or the b⁰+AT-encoding gene, *SLC7A9*, cause type A and type B autosomal recessive cystinuria, respectively.³⁴⁰ This disorder is characterized by renal cystine clearance that is close to GFR, whereas cationic amino acids remain partially reabsorbed. Cystinuria causes formation of kidney stones due to low

solubility of cystine.³⁴¹ In the PST, rBAT may have a different partner, namely the aspartate/glutamate transporter AGT1.³⁴² AGT1 is localized apically and operates as a cystine transporter, exchanging cystine for glutamate. Interestingly, this suggests that cystine is transported as a neutral amino acid in the PCT and as an anionic amino acid in the PST. In $b^0,+AT$ knockout mice fractional excretion of arginine reaches 80%, while only 11% of the tubular cystine load is excreted.³⁴³ This is consistent with the presence of additional cystine transporters, such as AGT1 and EAAT3 (see anionic amino acid transporters later). The rBAT protein is a highly glycosylated type II membrane protein that is connected by a disulfide bridge to the transporter subunit $b^0,+AT$ just outside the membrane.³⁴⁴ Heterodimer formation is essential for the exit of the complex from the endoplasmic reticulum.³⁴⁵ Transport experiments show that rbat/ $b^0,+AT$ is an obligatory antiporter that preferentially takes up cationic amino acids from the lumen in exchange for neutral amino acids.³⁴⁶ This directionality is imposed by the inside-negative cell membrane potential, and it is also confirmed by the lack of neutral amino acids in cystinuria.³⁴⁷ In mice, the AGT1 gene is only expressed in males and thus is unlikely to be essential for renal cystine reabsorption. In humans, isolated cystinuria, a rare disorder in which only cystine is elevated in urine, is caused by selected mutations in $b^0,+AT$ that affect the substrate selectivity of the transport subunit.³⁴⁸ Mutations in AGT1 have not been identified thus far.

Clinical Relevance: Transporters for Cationic Amino Acids and Cystine Apical Transporters

Urolithiasis occurs in most cases of cystinuria. The generation of kidney stones is managed through the combination of several treatments. Tiopronin (α -mercaptopropionylglycine) is administered to form adducts with cystine, which have a higher solubility than cystine. Potassium citrate is used to increase the pH of urine >7.5 . Reduction of animal protein intake and nocturnal fluid intake are recommended. All treatments are aimed to reduce cystine formation and precipitation.

BASOLATERAL TRANSPORTERS

A basolateral exit pathway for cationic amino acids is defined by the disease LPI,^{349,350} a rare disorder caused by mutations in the SLC7A7 light chain of the heteromeric transporter 4F2hc-SLC7A7.^{351,352} Plasma levels of cationic amino acids are reduced, which affects urea cycle function, causing adverse reaction to protein ingestion. Urine levels of cationic amino acids are very high, particularly that of lysine. Intestinal absorption of cationic amino acids is also affected. Although a functionally redundant transporter exists (4F2hc-SLC7A6), its expression is low and cannot replace malfunction of SLC7A7.³⁵³

The prevalence of exchangers for the transport of cationic and neutral amino acids sets these mechanisms apart from the paradigm set by glucose reabsorption. One possible reason could be the maintenance of cytosolic amino acid pools that are significantly higher than those observed in blood plasma and which are required for protein biosynthesis and amino acid homeostasis.³⁵⁴ In the presence

of basolateral facilitators, cytosolic amino acid pools would be close to those observed in the blood.

TRANSPORTERS FOR ANIONIC AMINO ACIDS

The main apical transporter for anionic amino acids in the proximal tubule is EAAT3 (named EAAC1 in rodents, SLC1A3),³⁵⁵ which transports both glutamate and aspartate.³⁵⁶ Mutations in EAAT3 cause the rare condition of dicarboxylic aminoaciduria,³⁵⁷ which is characterized by highly elevated urine levels of aspartate and glutamate that can reach or even exceed the filtered amounts. EAAT3 also contributes to cystine transport,³⁵⁸ but cystine is only marginally elevated in individuals with dicarboxylic aminoaciduria.³⁵⁷ In addition to the kidney, EAAT3 is expressed in neurons and the intestine and allows high cytosolic accumulation due to the cotransport of $3Na^+$ and $1H^+$ and the antiport of $1K^+$.³⁵⁹ Despite expression of EAAT3 in neurons, dicarboxylic aminoaciduria is considered a benign disorder,³⁶⁰ though an association of EAAT3 mutations with obsessive-compulsive disorder has been reported.³⁶¹ Expression of EAAT3 is relatively low in the PCT, increases toward the PST, and is also observed in the distal parts of the tubule.³⁵⁵ Nevertheless, 90% of the filtered anionic amino acid load is reabsorbed in the PCT. The transporter is regulated by osmolarity and amino acid deprivation.³⁶² A dedicated efflux pathway for glutamate in renal epithelial cells has not been identified. In fact, there is evidence of an accumulative glutamate transporter in the basolateral membrane, which would prevent efflux.³⁶³ In accordance, glutamate is intensively metabolized in epithelial cells and efflux may be limited. Moreover, members of the SLC22 family are nonspecific anion transporters, and those located in the basolateral membrane may serve as efflux pathways for glutamate.³⁶⁴

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QUESTION

1. What statement about the sodium glucose cotransporter SGLT2 in the kidney is incorrect?
 - a. SGLT2 reabsorbs >90% of the filtered glucose in an euglycemic individual with normal kidney function
 - b. SGLT2 is expressed in the early proximal tubule
 - c. SGLT2 inhibition enhances GFR in short term
 - d. SGLT2 inhibition excretes only ~50% of the filtered glucose in an euglycemic individual with normal kidney function
 - e. SGLT2 inhibition is uricosuric

Answer: c

Rationale: SGLT2 is expressed in the brush border of the early proximal tubule. SGLT2 reabsorbs >90% of the

filtered glucose in an euglycemic individual with normal kidney function. SGLT2 inhibition excretes only ~50% of the filtered glucose in an euglycemic individual with normal kidney function, because SGLT1 in the downstream later part of the proximal tubule can compensate when more glucose is delivered to that site. SGLT2 inhibition is uricosuric and can lower blood urate levels. SGLT2 inhibition reduces GFR in short term through the tubuloglomerular feedback mechanism and by increasing tubular back pressure. SGLT2 inhibition can preserve or maintain higher GFR in the long term in type 2 diabetic patients.